4MRNA User Manual

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[GitHub repository]

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1. Introduction

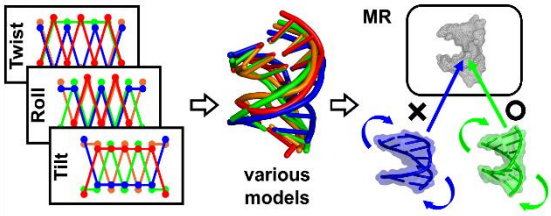
1-1. What is 4MRNA?

4MRNA is an abbreviation for "**M**assive **M**ulti-type **M**odel **M**olecular **R**eplacement for **N**ucleic **A**cids". This is a novel method designed to enhance molecular replacement (MR) for phasing in X-ray crystallography of nucleic acids.

Applying MR, which is widely used in structure determination (phase determination), to nucleic acids presents unique challenges that are not encountered with proteins. To overcome these issues, we developed an innovative strategy termed 4MRNA.

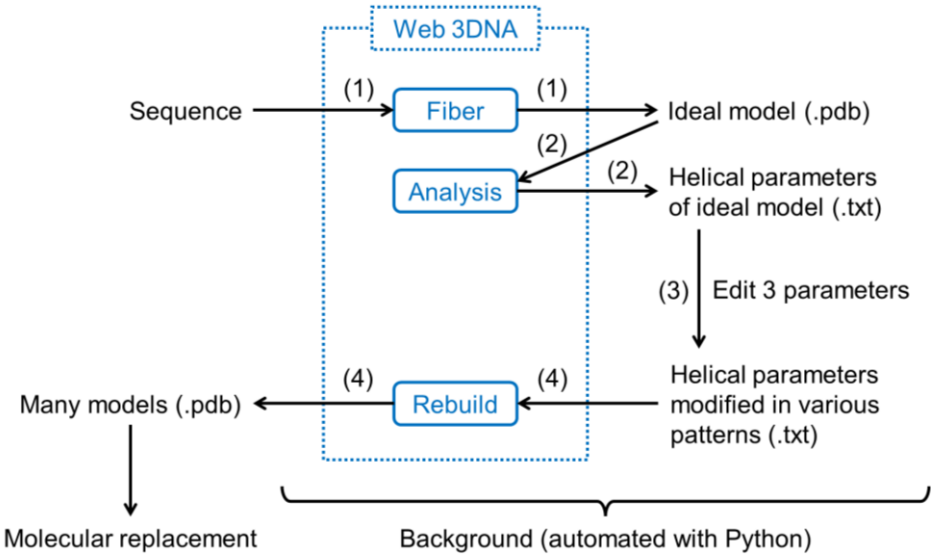
One major difficulty is that nucleic acids can adopt different 3D structures even with the same sequence. As a result, models existing in the database may not be suitable as search models, and sequence-based structure prediction methods such as AlphaFold are also limited in applicability [1,2]. Moreover, it is empirically known that MR of nucleic acids can fail even when the search model and the target structure differ only slightly. [3] To address these issues, the new strategy 4MRNA includes creating a large number of diverse search models (= massive multi-type models) and applying them to MR.

We found that by varying three out of the twelve parameters that control the three-dimensional structures of nucleic acids according to different patterns, the resulting set of models included ones that closely matched the correct structure, thereby increasing the success rate of MR.



(Strategy of 4MRNA)

Building on this strategy, we employed **Web 3DNA** [4], which is a web application that generates nucleic acid structural models based on parameters, to create a wide variety of models. The processes of parameter adjustment and model creation have been automated using **Python**. Subsequently, MR is carried out for each of the many created models. Since this operation must be repeated many times, we automated this process using **Shell** scripts on Linux.



(Tactics of 4MRNA)

1-2. Three ways to execute 4MRNA

There are three ways to execute 4MRNA: Standard style, Install style, and Customizable style.

In the **Standard style**, you simply run one Shell script named "**4MRNA.sh**" on your computer.

In the **Install style**, you install the `4MRNA` command on your computer in advance. Unlike Standard style, you do not need to download the Shell script every time. Instead, you can simply type the command `4MRNA` in Ubuntu/Terminal and run it. Thus, it is well suited for users who want to run the 4MRNA workflow frequently such as structural biologist and X-ray crystallographer.

In the **Customizable style**, you obtain some programming codes from the website called "**Programming Code Generator for 4MRNA**". Basically, you just run them in the prescribed order, but if needed, you can edit and customize the code. For example, you can adjust parameters by yourself to get more diverse models, or add models prepared by other methods to perform molecular replacement. Such customization represents a highly advanced option and is generally intended for experts in structural biology or X-ray crystallography.

Style	Procedure	Usability	Pre-setup	Customization	Target users
Standard	Run the code "4MRNA.sh"	Very good	Not required	Not possible	For all users
Install	Run the command `4MRNA`	Very good	Required	Not possible	For frequent users
Customizable	Download codes → Run them in order	Fair	Not required	Possible	For experts

For details on how to make the preparations, please refer to the Preparations section.

⇒ Preparations

For details on the usage of each style, please refer to the corresponding sections.

⇒ Standard style
⇒ Install style
⇒ Customizable style

2. Preparations

Section 2-1 describes the setup common to all three styles of 4MRNA. This preparation is required regardless of which style you use. Sections 2-2 to 2-4 describe the setup specific to each style, which should be carried out in addition to the common setup described in Section 2-1.

2-1. Common preparations for the three styles

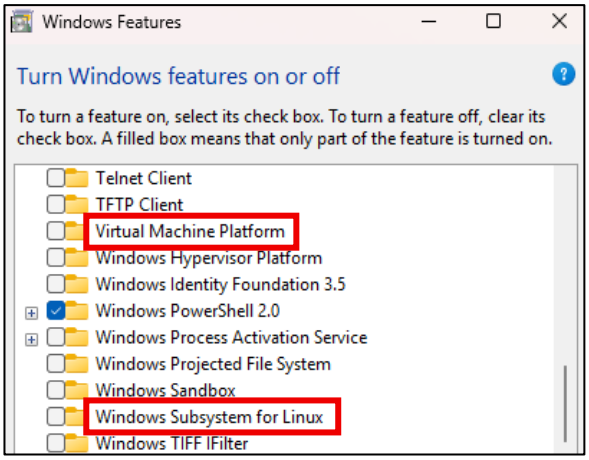
Please install and set up the following software in advance.

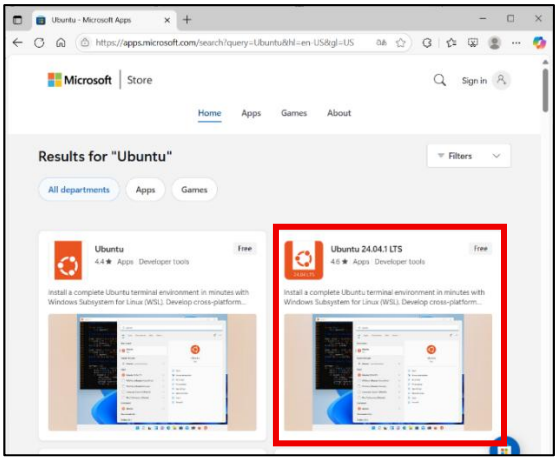
Name	Download link	Remarks
Ubuntu	https://apps.microsoft.com/search?query=Ubuntu	Required only for Windows to run Shell scripts. Not required for macOS or Linux, since they use Terminal.
Phaser	https://www.ccp4.ac.uk/download	The program Phaser is included in CCP4, so download and install CCP4.

2-1-1. Ubuntu

Step 1. Enable "**Windows Subsystem for Linux (WSL)**" and "**Virtual Machine Platform**". Go to *Control Panel* > *Programs* > *Programs and Features* > *Turn Windows features on or off*, and check the boxes for "*Windows Subsystem for Linux*" and "*Virtual Machine Platform*".

Step 2. Download Ubuntu from here.



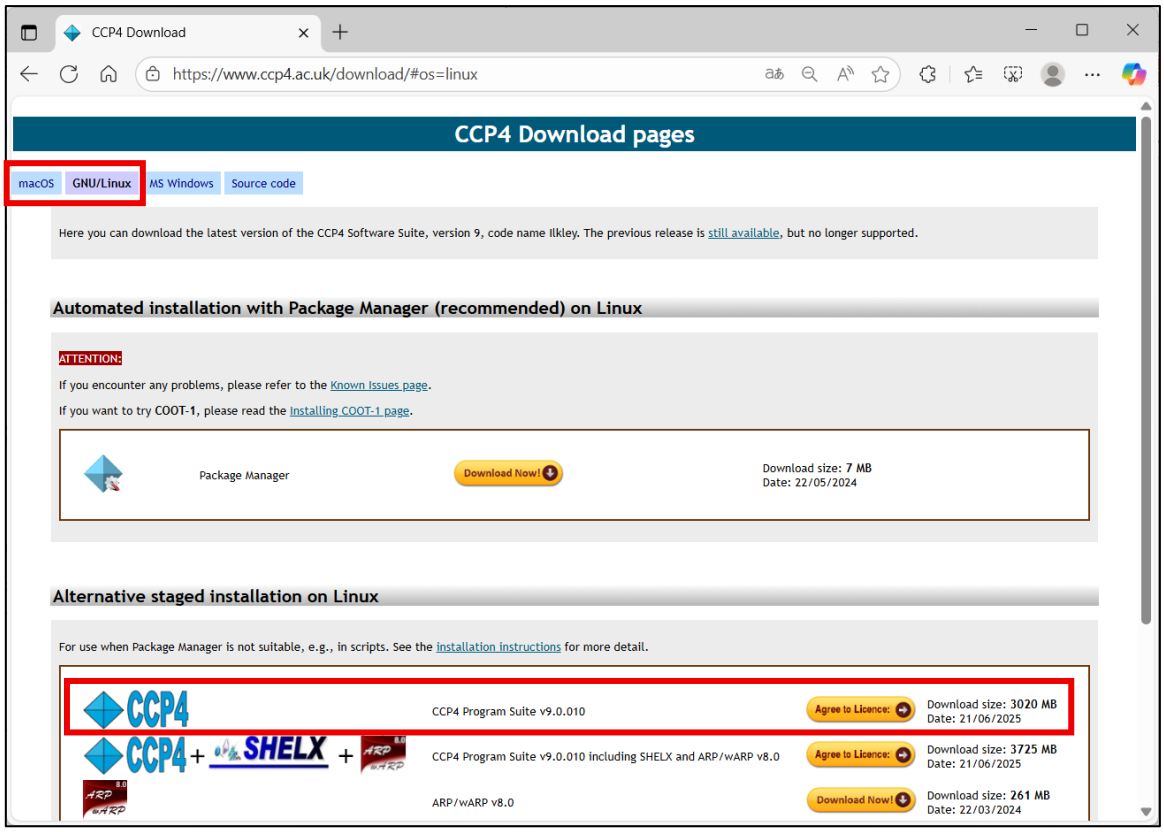


Step 3. After the download is complete, launch Ubuntu and run the following commands.

```
sudo apt update
sudo apt upgrade
sudo ln -s /usr/bin/python3 /usr/bin/python
sudo apt install python-pip
pip install --upgrade pip
```

2-1-2. Phaser

Step 1. Phaser is included in a software suite termed CCP4. Visit the official website and download the macOS or Linux version. Choose the Linux version even if you're using Windows.



Step 2. Once the download is complete, run the following commands in order to complete the installation. The following commands are for Ubuntu on Windows, but similar commands should be used for macOS and Linux. (These commands assume that the file “ccp4-9.0.010-linux64.tar.gz” has been downloaded to “C:\Users\name\Downloads”. You will need to adjust the version number and directory path as appropriate.)

```
sudo su
cd /usr/local
mv /mnt/c/Users/name/Downloads/ccp4-9.0.010-linux64.tar.gz .
gunzip ccp4-9.0.010-linux64.tar.gz
tar -xvf ccp4-9.0.010-linux64.tar
cd ccp4-9
./BINARY.setup
exit
echo "source /usr/local/ccp4-9/bin/ccp4.setup-sh" >> ~/.bashrc
source ~/.bashrc
```

2-2. Additional preparations for the Standard style

Launch Ubuntu or Terminal, and run the following commands.

```
pip install pandas playwright
playwright install
playwright install-deps
```

Note (added)

On newer versions of Ubuntu, the above commands may not work. In such cases, please create a virtual environment as shown below before proceeding with the setup.

```
mkdir ~/venvs
cd ~/venvs
python -m venv 4MRNA
pip install --upgrade pip
pip install pandas playwright
playwright install
playwright install-deps
```

Furthermore, if you have set up a virtual environment as described above, please run the following command immediately before running “4MRNA.sh”.

```
source ~/venvs/4MRNA/bin/activate
```

2-3. Additional preparations for the Install style

For Install style, the `4MRNA` command needs to be installed on your computer. Please run the following commands in order. If the version is displayed by running the final line of code, the installation is complete.

```
# Python setup
pip install pandas playwright
playwright install
playwright install-deps

# Installation of 4MRNA
cd $HOME
git clone https://github.com/S-Ando-Biophysics/4MRNA-Install.git
cd 4MRNA-Install
chmod +x install.sh
./install.sh
echo 'export PATH="$HOME/4MRNA-Install/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
4MRNA version
```

Note (added)

On newer versions of Ubuntu, the above commands in the Python setup section may not work. In such cases, please create a virtual environment as shown below before proceeding with the setup. The subsequent installation of 4MRNA can be performed as described without any changes.

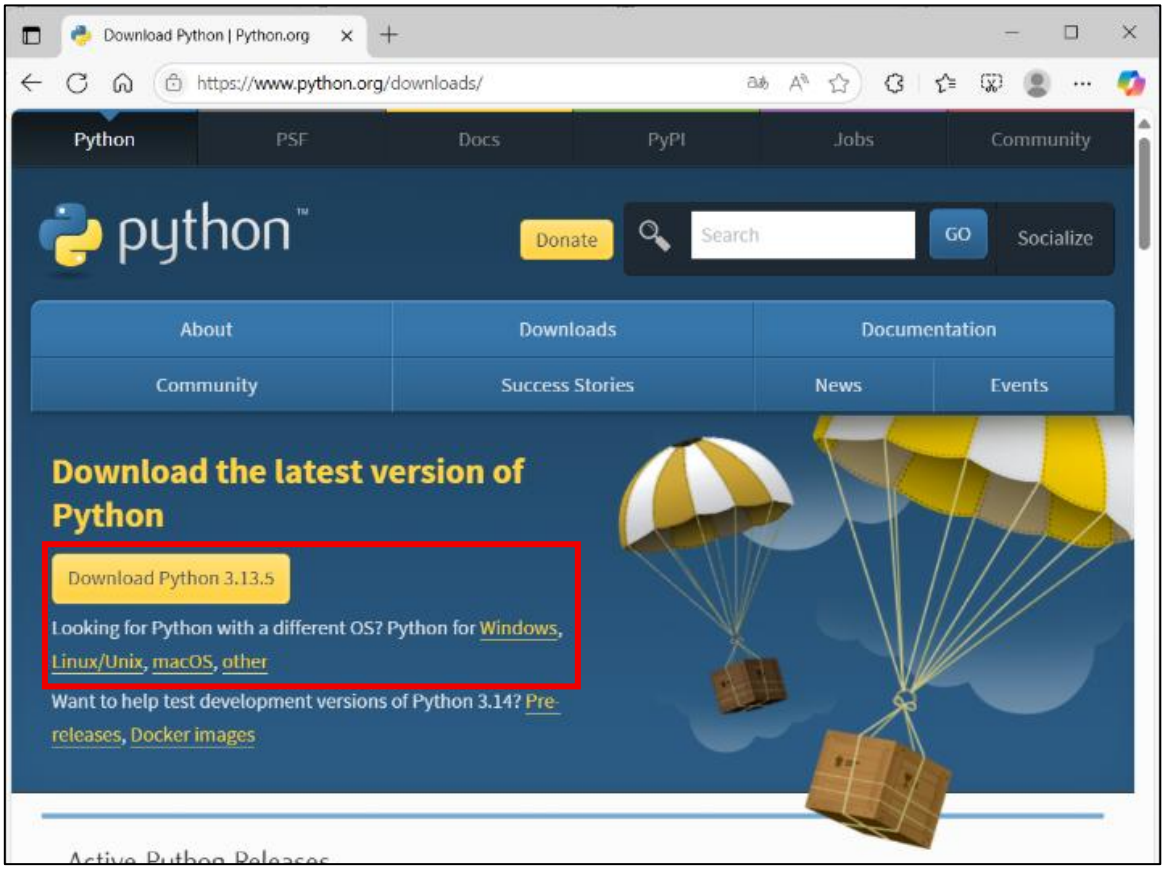
```
mkdir ~/venvs
cd ~/venvs
python -m venv 4MRNA
pip install --upgrade pip
pip install pandas playwright
playwright install
playwright install-deps
```

Furthermore, if you have set up a virtual environment as described above, please run the following command immediately before running the `4MRNA` command.

```
source ~/venvs/4MRNA/bin/activate
```

2-4. Additional preparations for the Customizable style

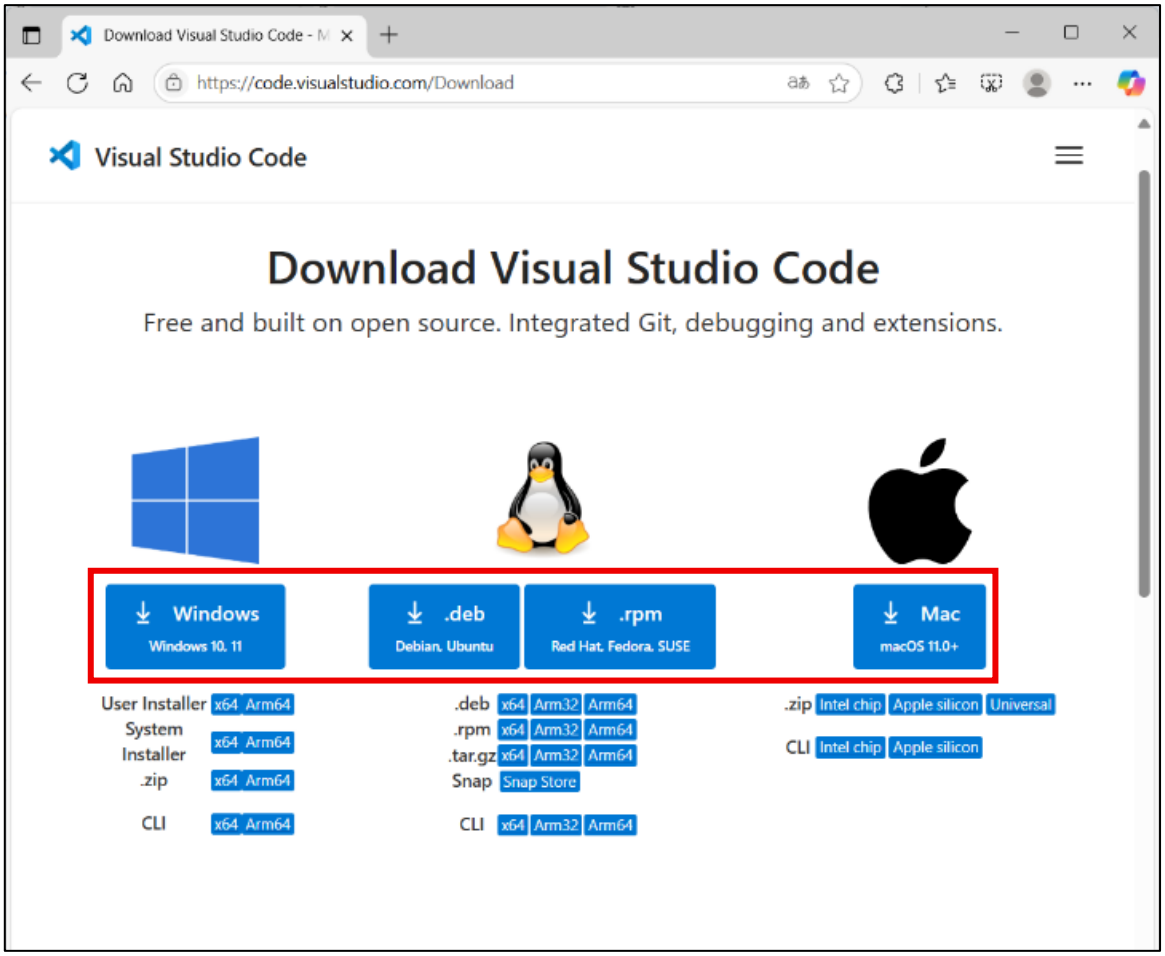
Step 1. Please install Python. Access the official website, download the installer suitable for your OS, and run it.



Step 2. After installation, open Terminal and execute the following command.

```
pip install pandas playwright
playwright install
```

Step 3. It is recommended (though not mandatory) to download a text editor such as Visual Studio Code, which makes it easier to edit or run Python codes.



2-5. Notes

- **[For macOS]** If no browsers other than Safari (such as Google Chrome, Microsoft Edge, or Firefox) are installed, the following additional setting may be required. Go to *Safari* > *Preferences* > *Develop*, and check "*Allow Remote Automation*".
- **[For macOS and Linux]** Although Python is often preinstalled, it may be missing, outdated, or the `pip` command may not be available. In such cases, please refer to Section 2-4 and Step2 of Section 2-1-1 to set up your environment.

3. How to use the Standard style

3-1. Quick summary of the procedure

Step 1. Create a new directory on your computer for executing 4MRNA. After creating it, place your reflection file (.mtz) for molecular replacement into that directory.

Step 2. Download "**4MRNA.sh**" into the directory created in Step 1, either directly from the following URL or via the following website.

- **Directly**: https://github.com/S-Ando-Biophysics/4MRNA/releases/latest/download/4MRNA.sh
- **Website**: https://s-ando-biophysics.github.io/4MRNA/

Step 3. Open Ubuntu/Terminal, change to the directory created in Step 1, and run "**4MRNA.sh**" as follows.

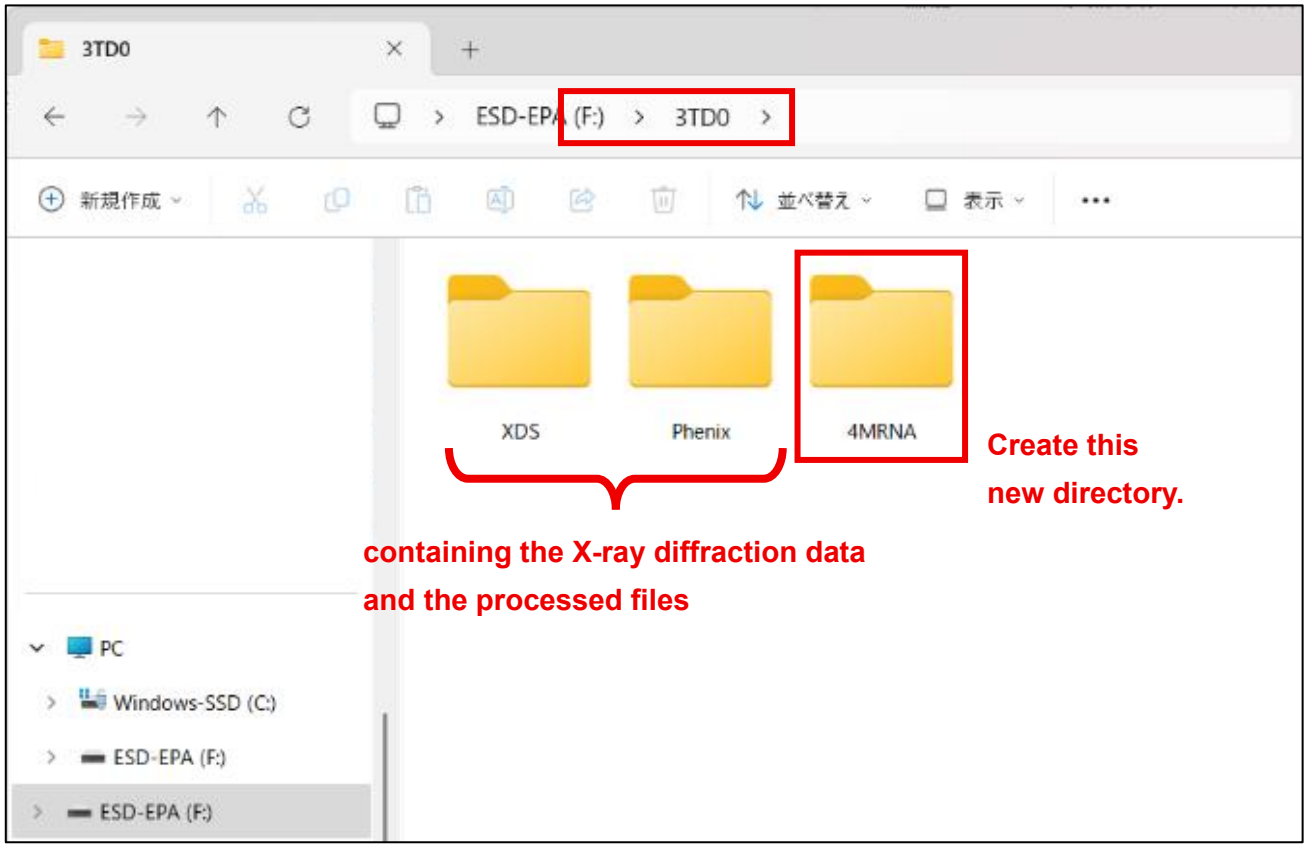
`bash 4MRNA.sh` or `chmod +x 4MRNA.sh && ./4MRNA.sh`

Step 4. When 4MRNA finishes, a directory named "**4MRNA-Results**" will be created, containing up to seven candidate solutions for molecular replacement.

3-2. Step-by-step instructions with screenshots

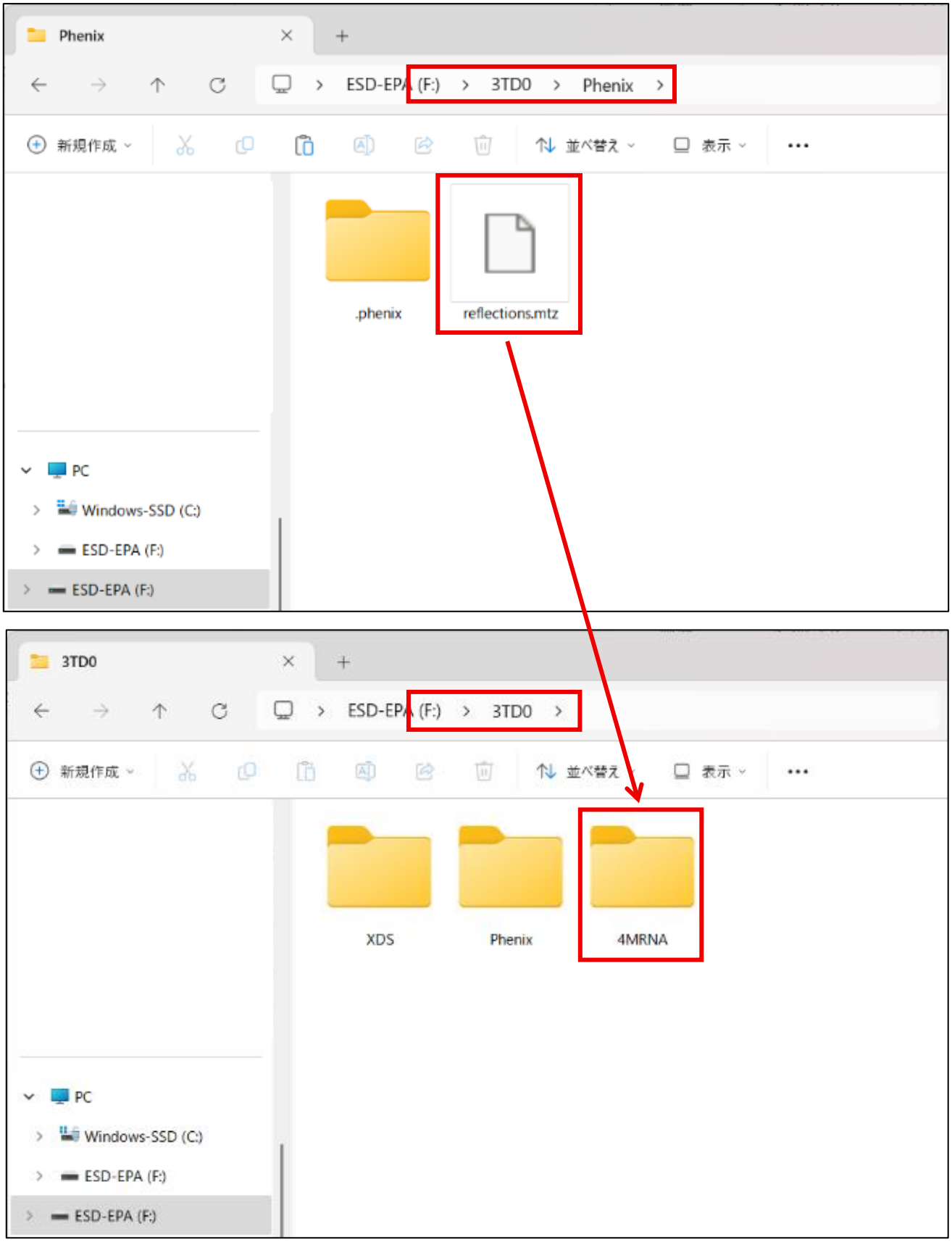
■ Create a new directory on your computer for executing 4MRNA.

[e.g.] Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a directory 3TD0 on a USB drive (F:). In that case, create a new directory named "4MRNA" inside "F:\3TD0".

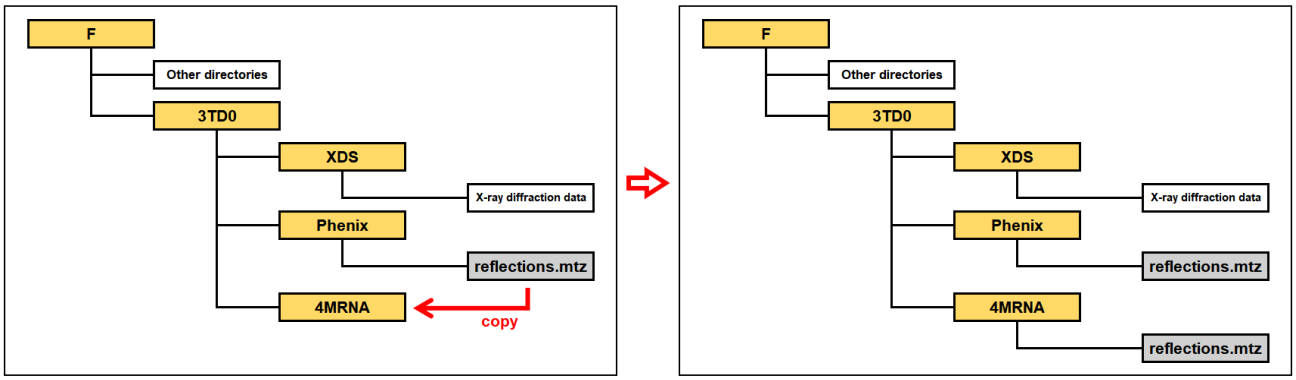


■ Place your reflection file (.mtz) for molecular replacement into the created directory.

[e.g.] When using the software Phenix, the reflection file is typically named “reflections.mtz”. If this file is in “F:\3TD0\Phenix”, copy it, and paste to “F:\3TD0\4MRNA”.

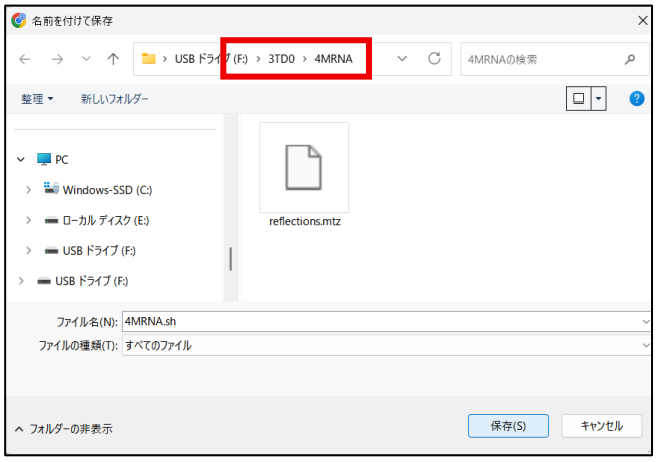


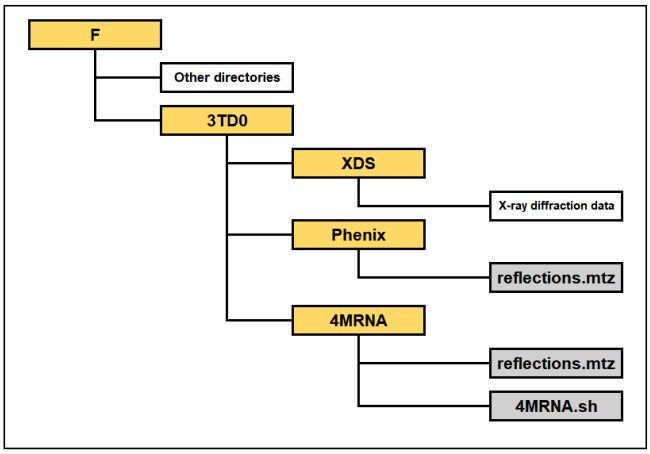
■ The directory structure after these operations is as illustrated in the example below.



■ Download "**4MRNA.sh**" directly from the following URL into the created directory.
https://github.com/S-Ando-Biophysics/4MRNA/releases/latest/download/4MRNA.sh

[e.g.] If the path of the created directory is "F:\3TD0\4MRNA", download "4MRNA.sh" there. After downloading, the directory structure will appear as in the example figure on the right below.





■ Open Ubuntu or Terminal, change to the created directory including "4MRNA.sh", and run it as follows. Only if you use the virtual environment, please type `source ~/venvs/4MRNA/bin/activate` immediately before running "4MRNA.sh". (For details, please refer to Section 2-2.)

`bash 4MRNA.sh` or `chmod +x 4MRNA.sh && ./4MRNA.sh`

[e.g.] If the created directory is "F:\3TD0\4MRNA" and it contains "4MRNA.sh", enter the command in Ubuntu as shown in the figure below. (This example is for Windows.)

- Actually, for “4MRNA.sh” to work, an input file called “**4MRNA-INP.txt**” is required. The user is prompted to choose from two options: (1) preparing the file yourself (a template will be provided), or (2) answering questions on the screen so that the program automatically generates it.
- Option **(2)** is recommended. For option (1), refer to the supplementary notes in Section 3-3-2.
- If you choose option (2), several questions will be asked. Please answer them. Once all questions have been answered, the 4MRNA workflow will start automatically.

[e.g.] The figure below shows an example of molecular replacement using two different RNAs as models. In this example, during molecular replacement, one copy of the first model and two copies of the second model are searched for in the asymmetric unit of the crystal.

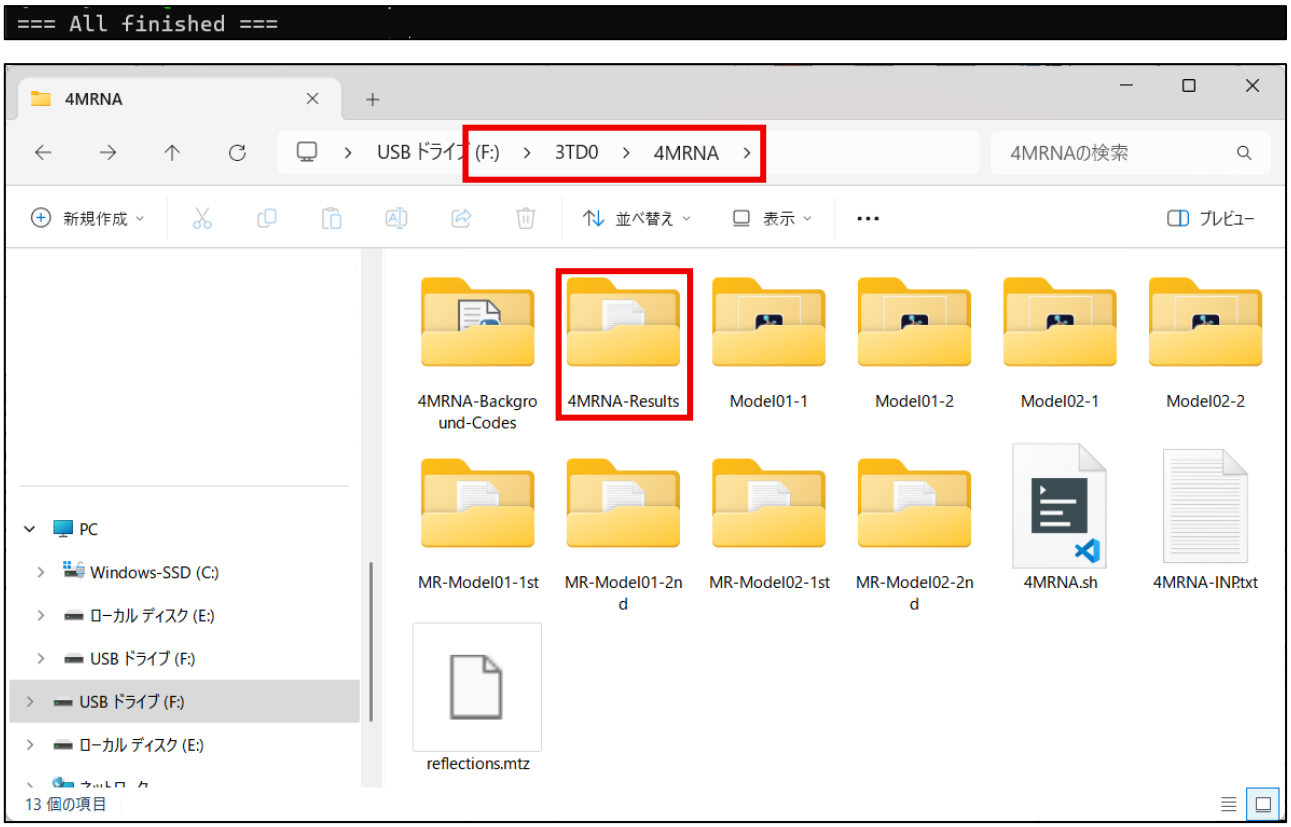
```
Input file '4MRNA-INP.txt' not found.
Choose one of the following:
  1) I will prepare it myself. (A template will be provided.)
  2) I will answer questions here to generate it now.
Select [1/2]: 2
How many models for molecular replacement? [1,2,3]: 2

Enter information of model #1.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): CGUCGACG
How many copies of the model to be searched in molecular replacement: 1

Confirm model #1:
-TYPE1 A-RNA
-SEQ1 CGUCGACG
-NUM1 1
Is this correct? [Y/N]: Y

Enter information of model #2.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): GCGUC
How many copies of the model to be searched in molecular replacement: 2

Confirm model #2:
-TYPE2 A-RNA
-SEQ2 GCGUC
-NUM2 2
Is this correct? [Y/N]: Y
```

- The figures below provide examples of the execution screen.
- Initially, the contents of the input file (i.e., the answers to the questions) are read, and settings required for running 4MRNA, such as calculating the molecular weight, are carried out.
- Once the initial setup is complete, “Start running” will be displayed.

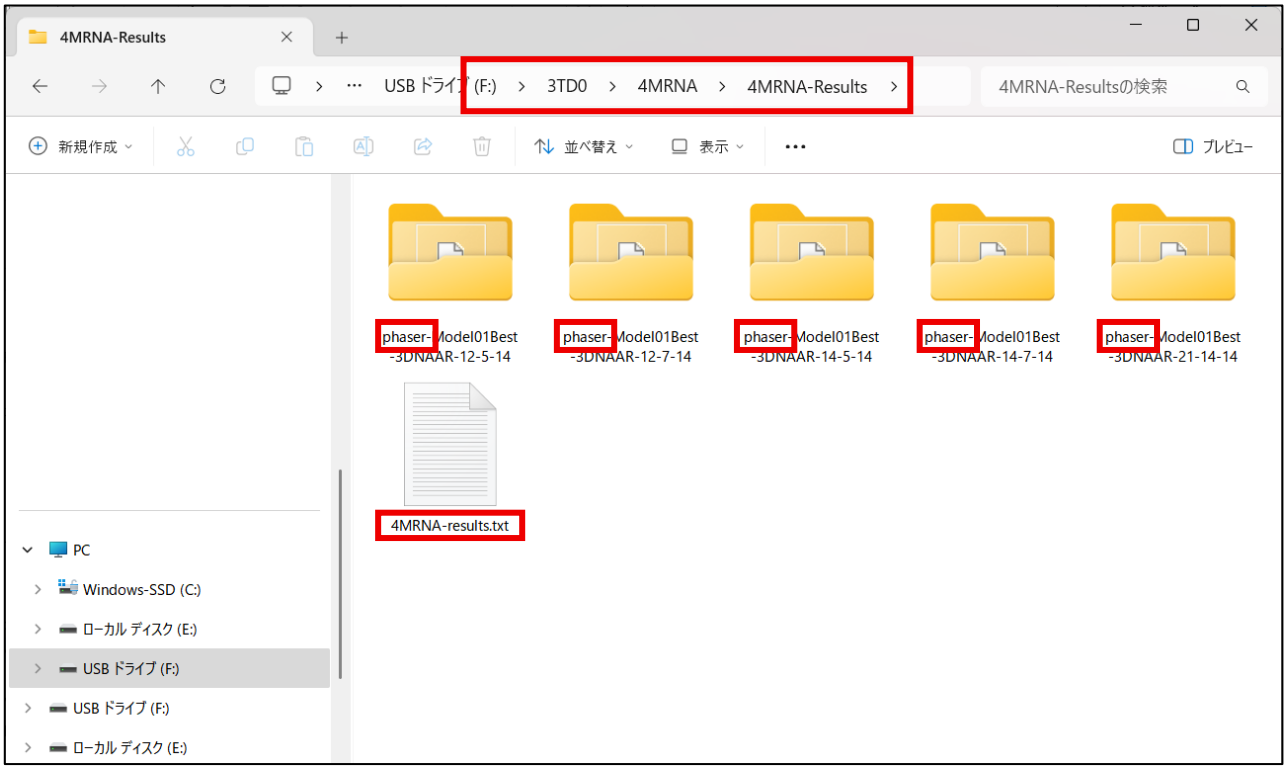
```
Continuing…
Detected model types: 2
  Model01: TYPE=A-RNA  SEQ=CGUCGACG  NUM=1  MW=5051.38
  Model02: TYPE=A-RNA  SEQ=GCGUC  NUM=2  MW=3115.14
```

```
=== Start running (in /mnt/f/3TD0/4MRNA/4MRNA-Background-Codes) ===
```

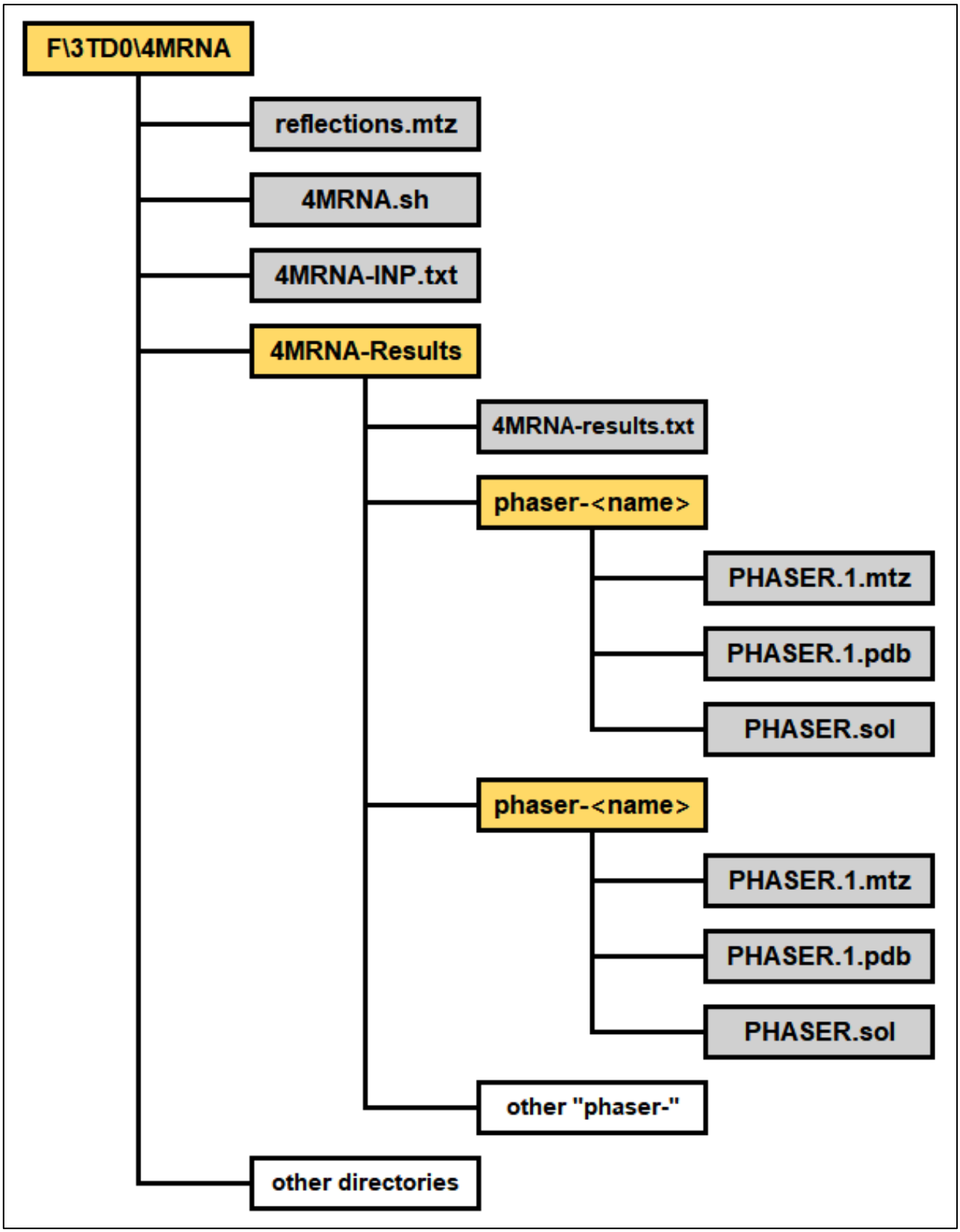
■ When the entire workflow is finished, the message “All finished” will be displayed, and a directory named “**4MRNA-Results**” will be created inside the 4MRNA execution directory (i.e. the directory you created at the beginning).



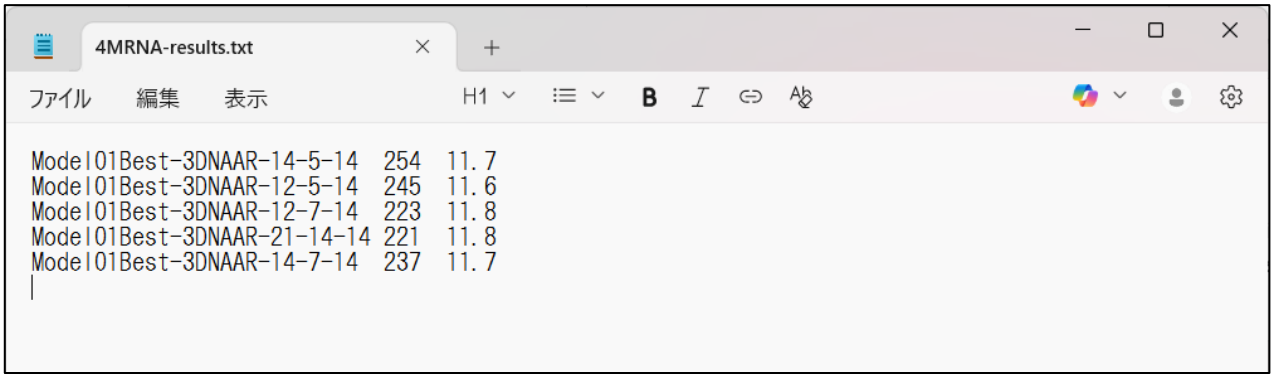
■ Inside the “4MRNA-Results” directory, there is a file named “**4MRNA-results.txt**” and up to seven subdirectories. Each subdirectory name begins with “**phaser-**” and contains the results of molecular replacement (MR).



- Inside each “phaser-” subdirectory, there is a structural file named “**PHASER.1.pdb**.” This file is a solution of molecular replacement, that is, a candidate for the correct structure.
- The following figure is an example of the final directory structure.



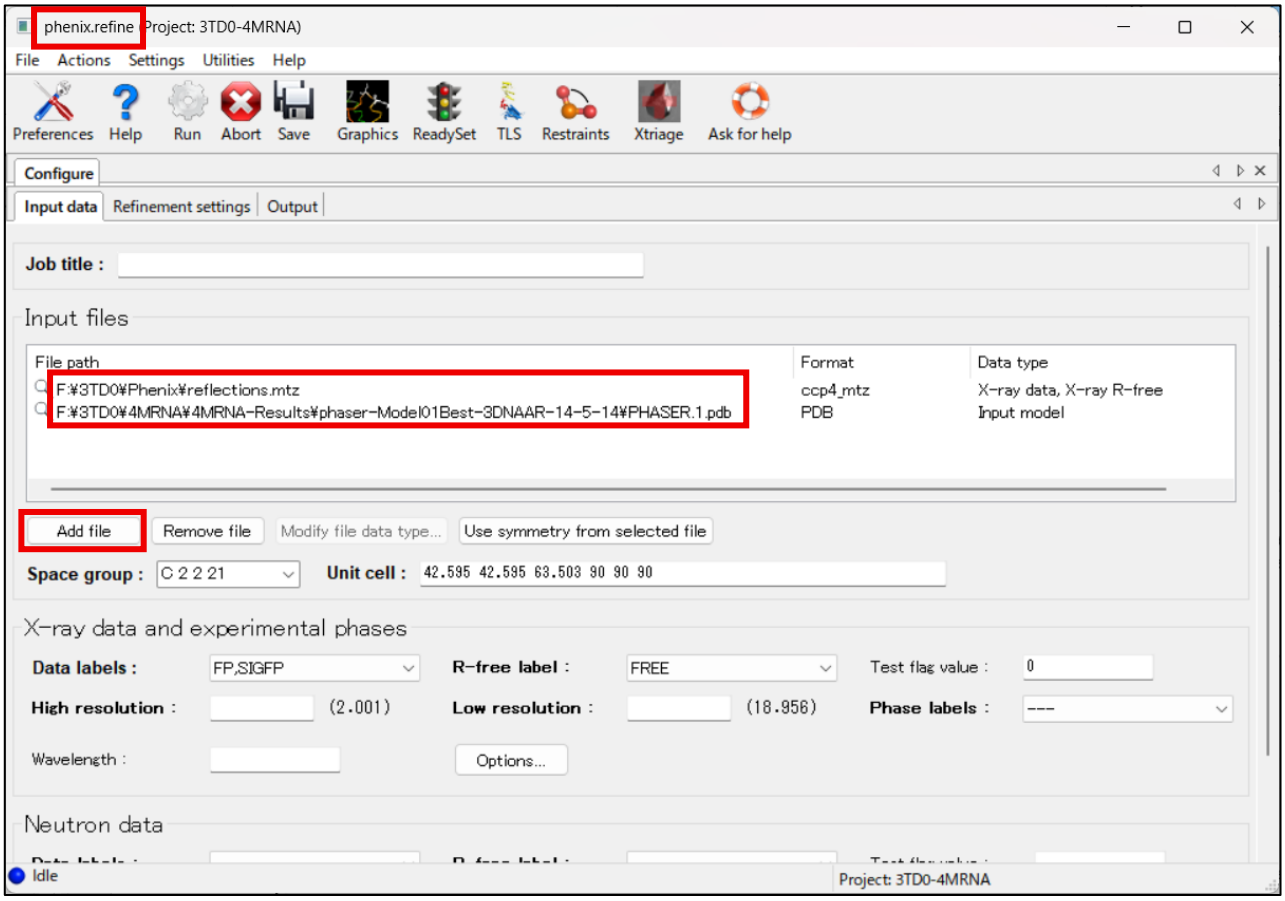
■ Each run of molecular replacement produces a solution, but it must be validated against the experimental data (electron density) to confirm its correctness. This validation is guided by two statistical values, **LLG** and **TFZ**. The program is designed to automatically select up to seven solutions with the most favorable statistics. These statistics are compiled in "**4MRNA-results.txt**." Each line in this file lists the ID of molecular replacement (i.e. the name of directory) , LLG, and TFZ in that order.



■ With reference to the statistics provided in "**4MRNA-results.txt**," open the structural file ("**PHASER.1.pdb**") together with the electron density map ("**PHASER.1.mtz**") in software like Coot. By overlaying them, decide which of the up to seven candidate solutions is the best.

■ After the optimal structure has been determined, it will be used as the initial structure for refinement.

[e.g.] For example, if the structure with the molecular replacement ID "Model01Best-3DNAAR-14-5-14" (i.e., "PHASER.1.pdb" within the directory "phaser-Model01Best-3DNAAR-14-5-14") is decided as the initial structure, you should add this file. In addition, add the reflection file ("reflections.mtz").

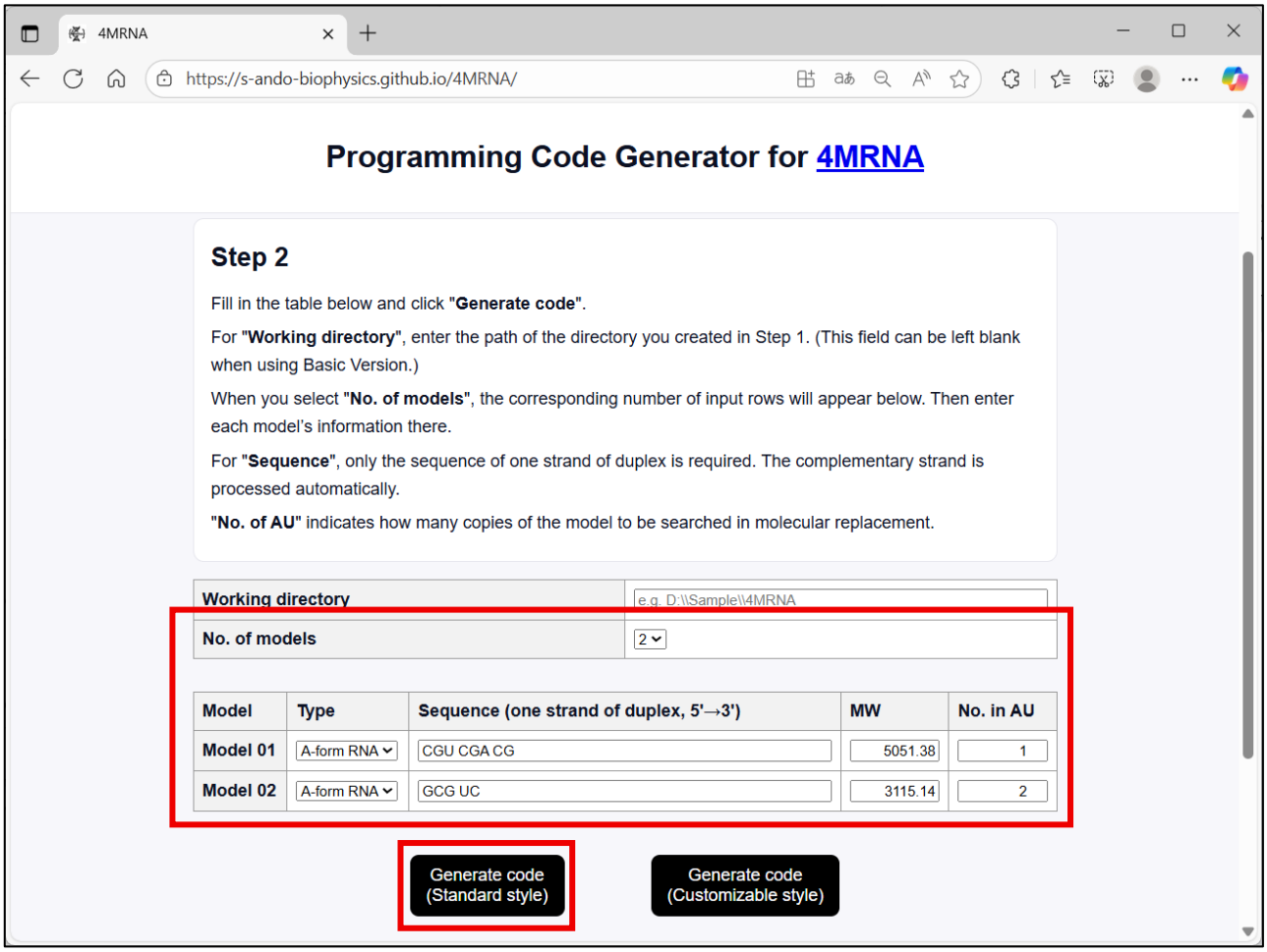


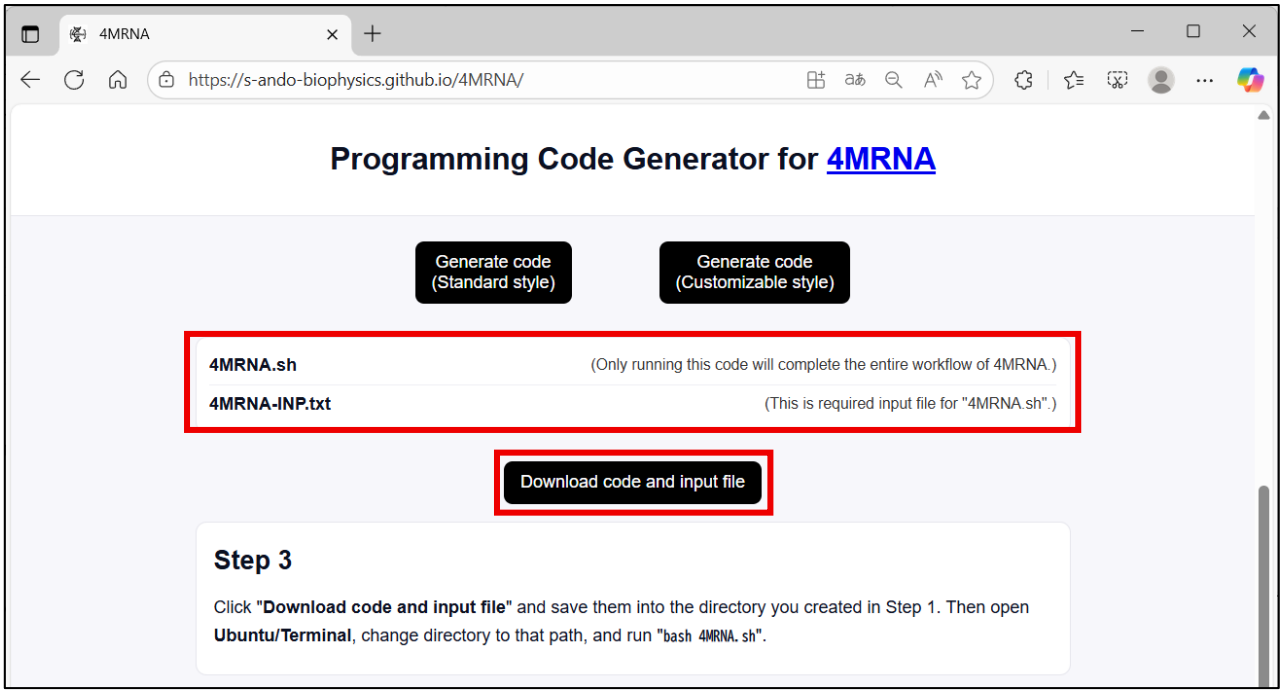
3-3. Supplementary notes

3-3-1. Download “4MRNA.sh” from the website

- In the step-by-step description in Section 3-2, “4MRNA.sh” was obtained directly from the URL. However, it can also be downloaded from the following website called “**Programming Code Generator for 4MRNA**”.
  https://s-ando-biophysics.github.io/4MRNA/
- Once you access the website, enter the required information in the table according to the instructions provided, and then click “**Generate code (Standard style)**” button.
- Clicking “Generate code (Standard style)” generates two files: “**4MRNA.sh**” and the input file “**4MRNA-INP.txt**”. You can download them by clicking “**Download code and input file**” button. Please make sure to download them into the directory you created at the beginning of the procedure.
- The generated input file “4MRNA-INP.txt” is already completed, containing the information entered in the table.
- All you need to do is run “4MRNA.sh.” As described in the step-by-step instructions in Section 3-2, open Ubuntu or Terminal, change to the created directory including “4MRNA.sh”, and run it as follows.

`bash 4MRNA.sh` or `chmod +x 4MRNA.sh && ./4MRNA.sh`





3-3-2. About the input file "4MRNA-INP.txt"

- Actually, for "4MRNA.sh" to work, an input file called "**4MRNA-INP.txt**" is required.
- There are three ways to prepare the input file "4MRNA-INP.txt":
  1. Prepare the input file yourself before running "4MRNA.sh".
     ⇒ Please refer to the template below.
  2. By downloading "4MRNA.sh" from the website ("Programming Code Generator for 4MRNA"), which simultaneously provides a completed version of the input file.
     ⇒ Please refer to Section 3-3-1.
  3. Do not prepare the input file in advance. Instead, answer the questions prompted by "4MRNA.sh", and the program will automatically create the input file.
     ⇒ Please refer to Section 3-2.
- The template of the input file "4MRNA-INP.txt" looks like the screenshot below. The following three pieces of information should be repeated for each model used in molecular replacement.

  -TYPE: This is the nucleic acid type. Choose from A-form DNA, B-form DNA, or A-form RNA.

  -SEQ: This is the sequence. Only the sequence of one strand of the duplex is required. The complementary strand is processed automatically by the program.

  -NUM: This means how many copies of the model to be searched in molecular replacement.

```
-TYPE1 A-DNA
-SEQ1 ACGTACGTACGT
-NUM1 1

-TYPE2 B-DNA
-SEQ2 ACGTACGT
-NUM2 1

-TYPE3 A-RNA
-SEQ3 ACGU
-NUM3 1
```

4. How to use the Install style

In this style, you install the `4MRNA` command on your computer in advance. Installation is, of course, required only once at the initial use. This Section 4 describes the usage of the `4MRNA` command after installation has been completed. For instructions on how to install, please refer to Section 2-2.

4-1. Quick summary of the procedure

Step 1. Create a new directory on your computer for executing 4MRNA. After creating it, place your reflection file (.mtz) for molecular replacement into that directory.

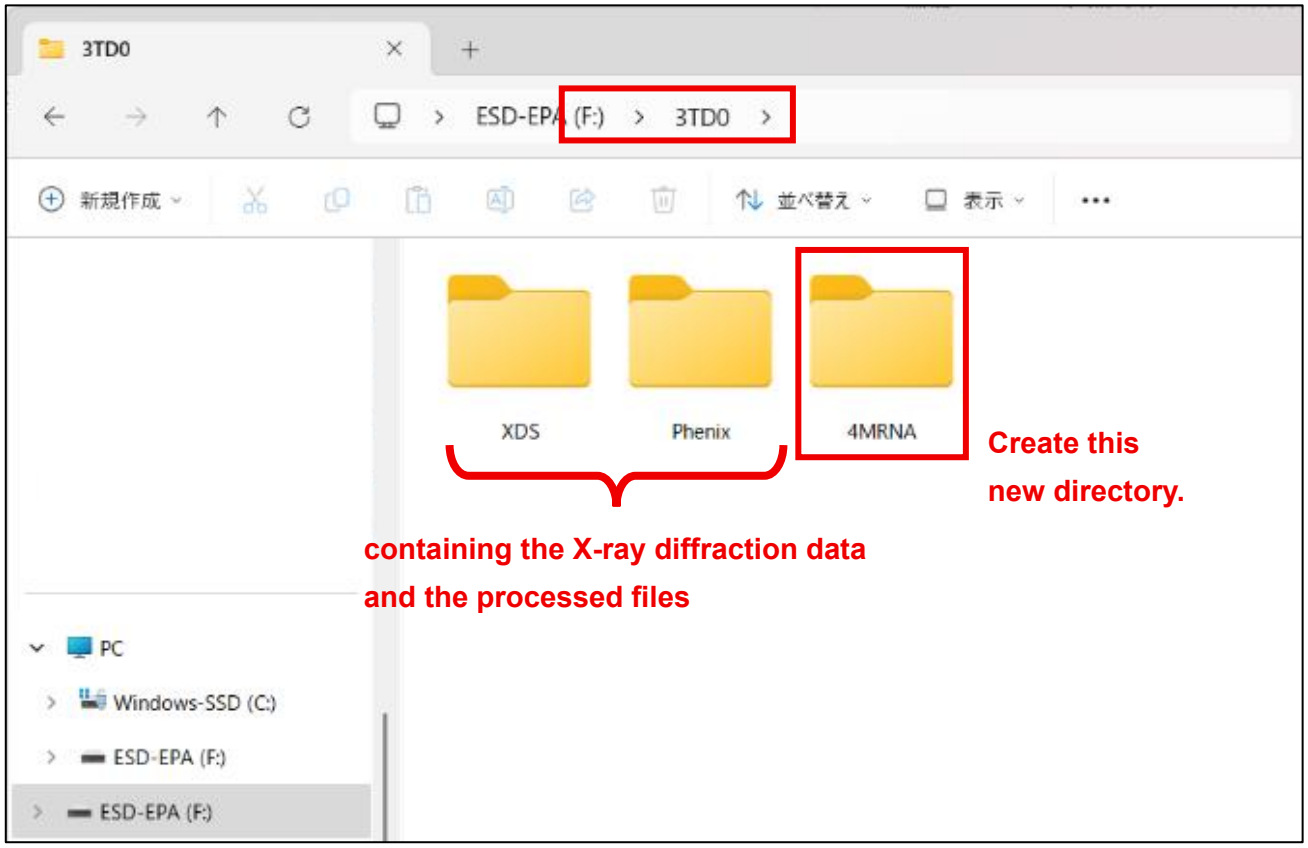
Step 2. Open Ubuntu/Terminal, change to the directory created in Step 1, and run the command `4MRNA`.

Step 3. When 4MRNA finishes, a directory named "**4MRNA-Results**" will be created, containing up to seven candidate solutions for molecular replacement.

4-2. Step-by-step instructions with screenshots

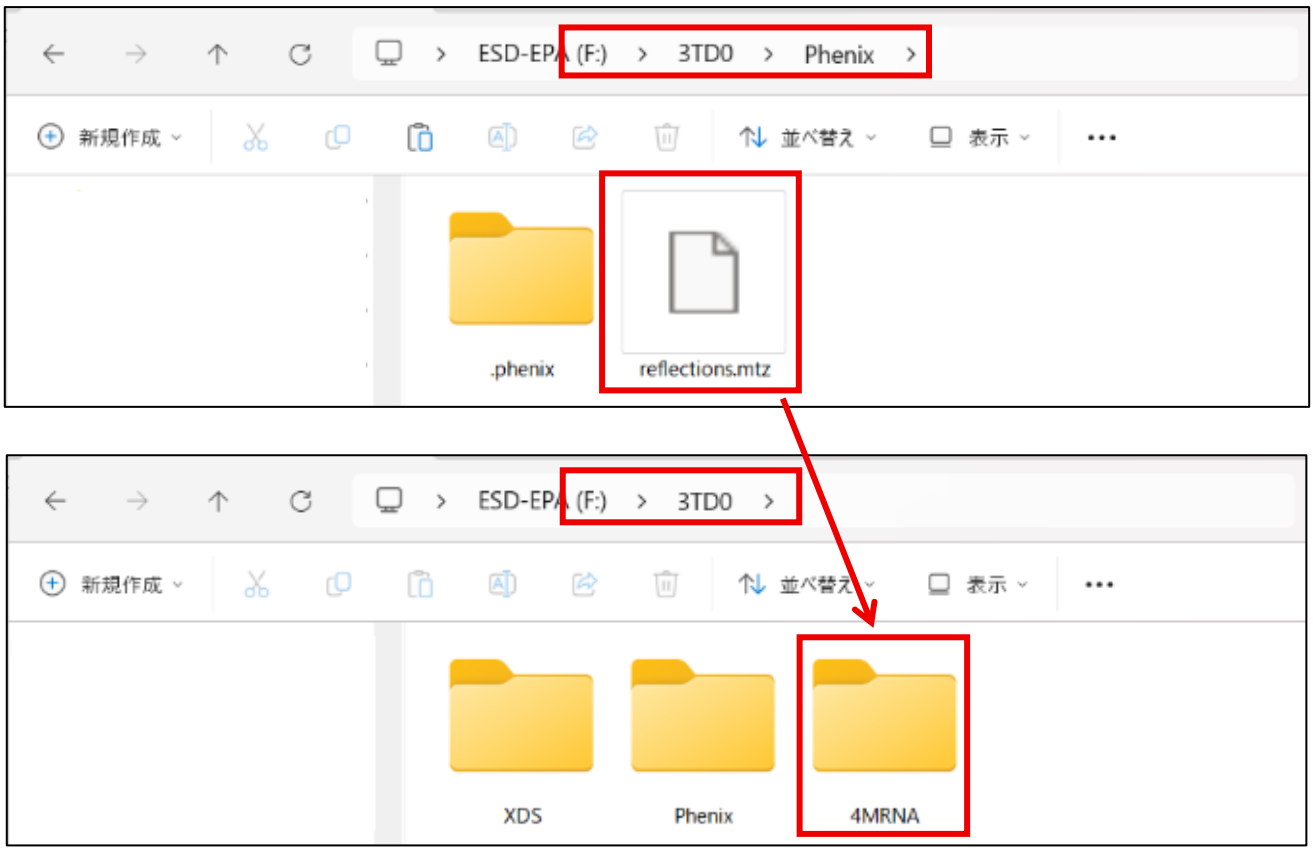
■ Create a new directory on your computer for executing 4MRNA.

[e.g.] Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a directory 3TD0 on a USB drive (F:). In that case, create a new directory named "4MRNA" inside "F:\3TD0".

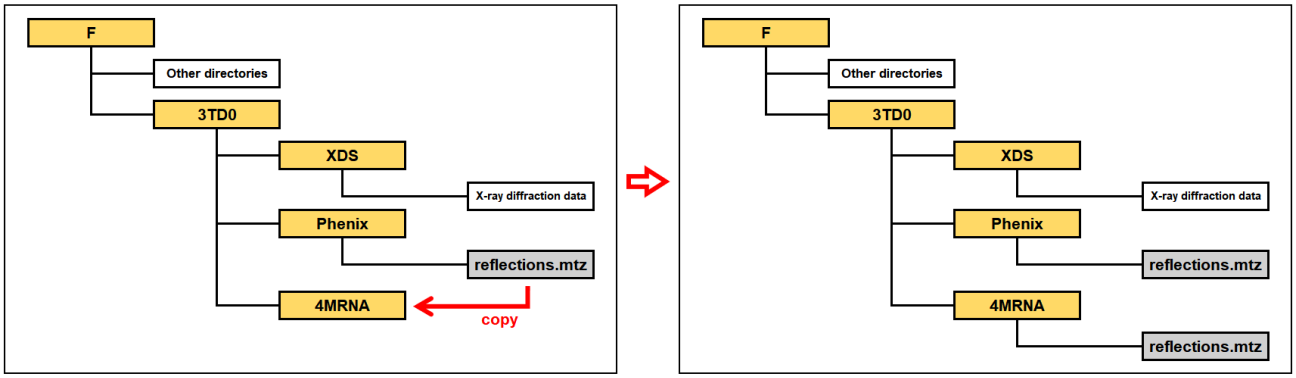


■ Place your reflection file (.mtz) for molecular replacement into the created directory.

[e.g.] When using the software Phenix, the reflection file is typically named “reflections.mtz”. If this file is in “F:\3TD0\Phenix”, copy it, and paste to “F:\3TD0\4MRNA”.

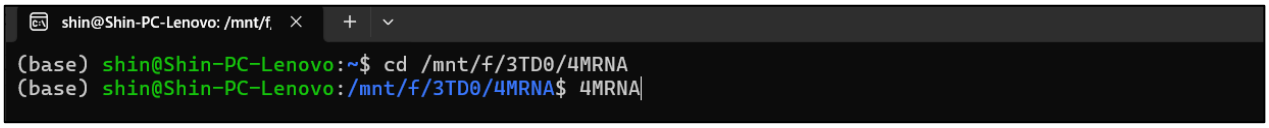


■ The directory structure after these operations is as illustrated in the example below.



■ Open Ubuntu or Terminal, change to the created directory, and run the command `4MRNA` as follows. Only if you use the virtual environment, please type `source ~/venvs/4MRNA/bin/activate` immediately before running `4MRNA`. (For details, please refer to Section 2-3.)

[e.g.] If the created directory is “F:\3TD0\4MRNA”, enter the command in Ubuntu as shown in the figure below. (This example is for Windows.)



- Actually, for the command **4MRNA** to work, an input file called "**4MRNA-INP.txt**" is required. The user is prompted to choose from two options: (1) preparing the file yourself (a template will be provided), or (2) answering questions on the screen so that the program automatically generates it.
- Option **(2)** is strongly recommended. For option (1), refer to the supplementary notes in Section 4-3.
- If you choose option (2), several questions will be asked. Please answer them. Once all questions have been answered, the 4MRNA workflow will start automatically.

[e.g.] The figure below shows an example of molecular replacement using two different RNAs as models. In this example, during molecular replacement, one copy of the first model and two copies of the second model are searched for in the asymmetric unit of the crystal.

```
Input file '4MRNA-INP.txt' not found.
Choose one of the following:
  1) I will prepare it myself. (A template will be provided.)
  2) I will answer questions here to generate it now.
Select [1/2]: 2
How many models for molecular replacement? [1,2,3]: 2

Enter information of model #1.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): CGUCGACG
How many copies of the model to be searched in molecular replacement: 1

Confirm model #1:
-TYPE1 A-RNA
-SEQ1 CGUCGACG
-NUM1 1
Is this correct? [Y/N]: Y

Enter information of model #2.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): GCGUC
How many copies of the model to be searched in molecular replacement: 2

Confirm model #2:
-TYPE2 A-RNA
-SEQ2 GCGUC
-NUM2 2
Is this correct? [Y/N]: Y
```

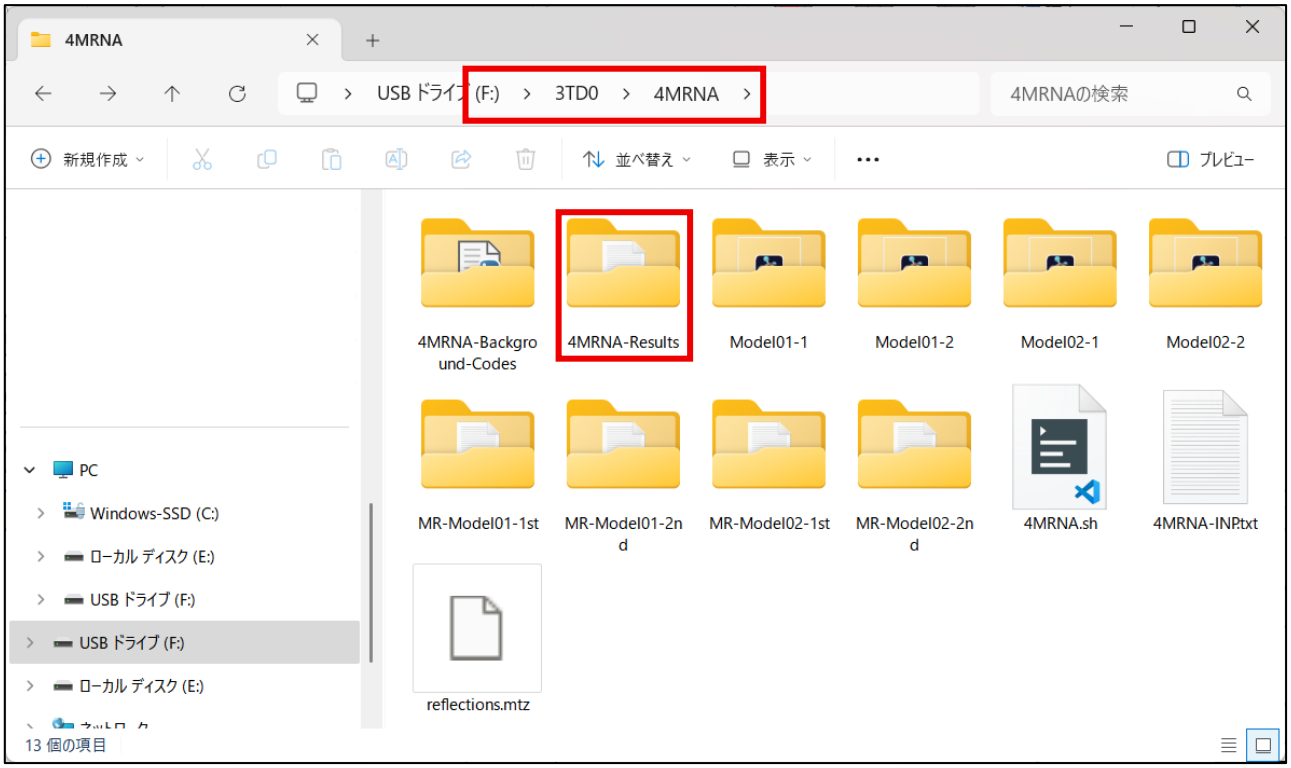
- The figures below provide examples of the execution screen.
- Initially, the contents of the input file (i.e., the answers to the questions) are read, and settings required for running 4MRNA, such as calculating the molecular weight, are carried out.
- Once the initial setup is complete, "Start running" will be displayed.

```
Continuing…
Detected model types: 2
  Model01: TYPE=A-RNA  SEQ=CGUCGACG  NUM=1  MW=5051.38
  Model02: TYPE=A-RNA  SEQ=GCGUC  NUM=2  MW=3115.14
```

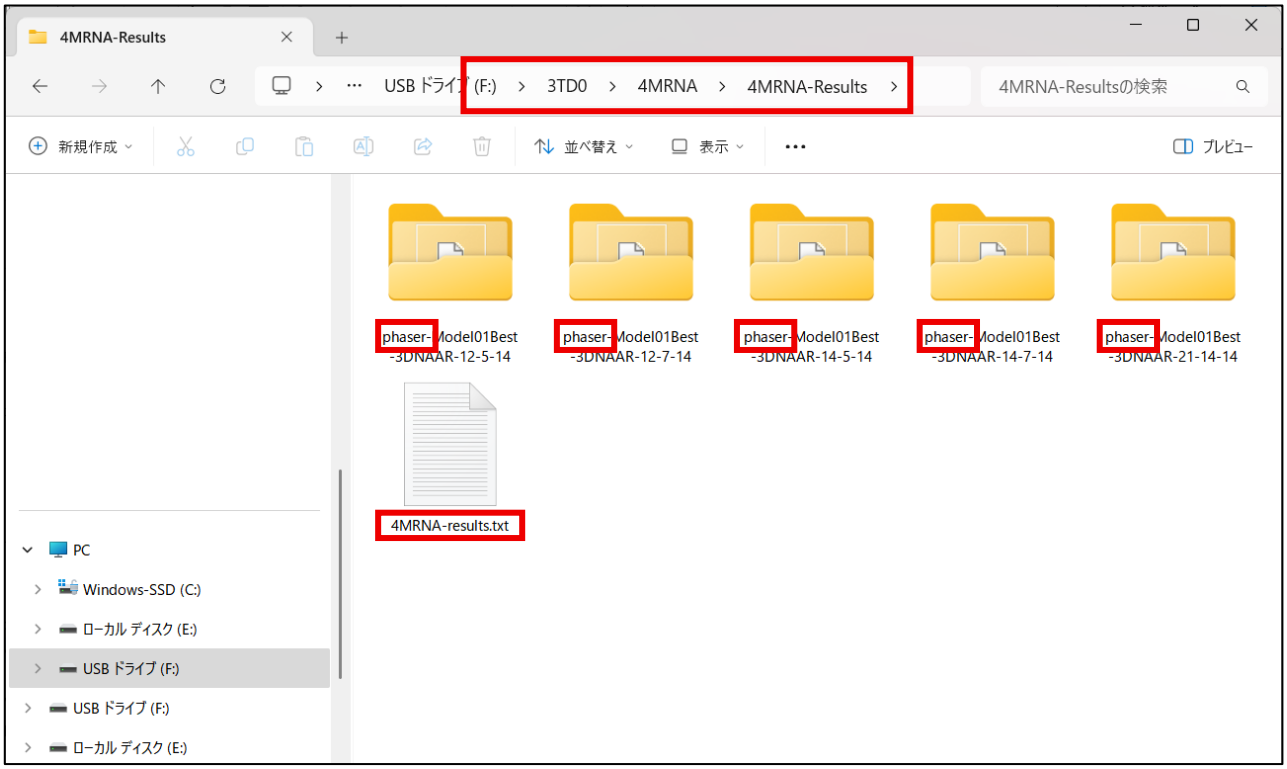
```
=== Start running (in /mnt/f/3TD0/4MRNA/4MRNA-Background-Codes) ===
```

■ When the entire workflow is finished, the message “All finished” will be displayed, and a directory named “**4MRNA-Results**” will be created inside the 4MRNA execution directory (i.e. the directory you created at the beginning).

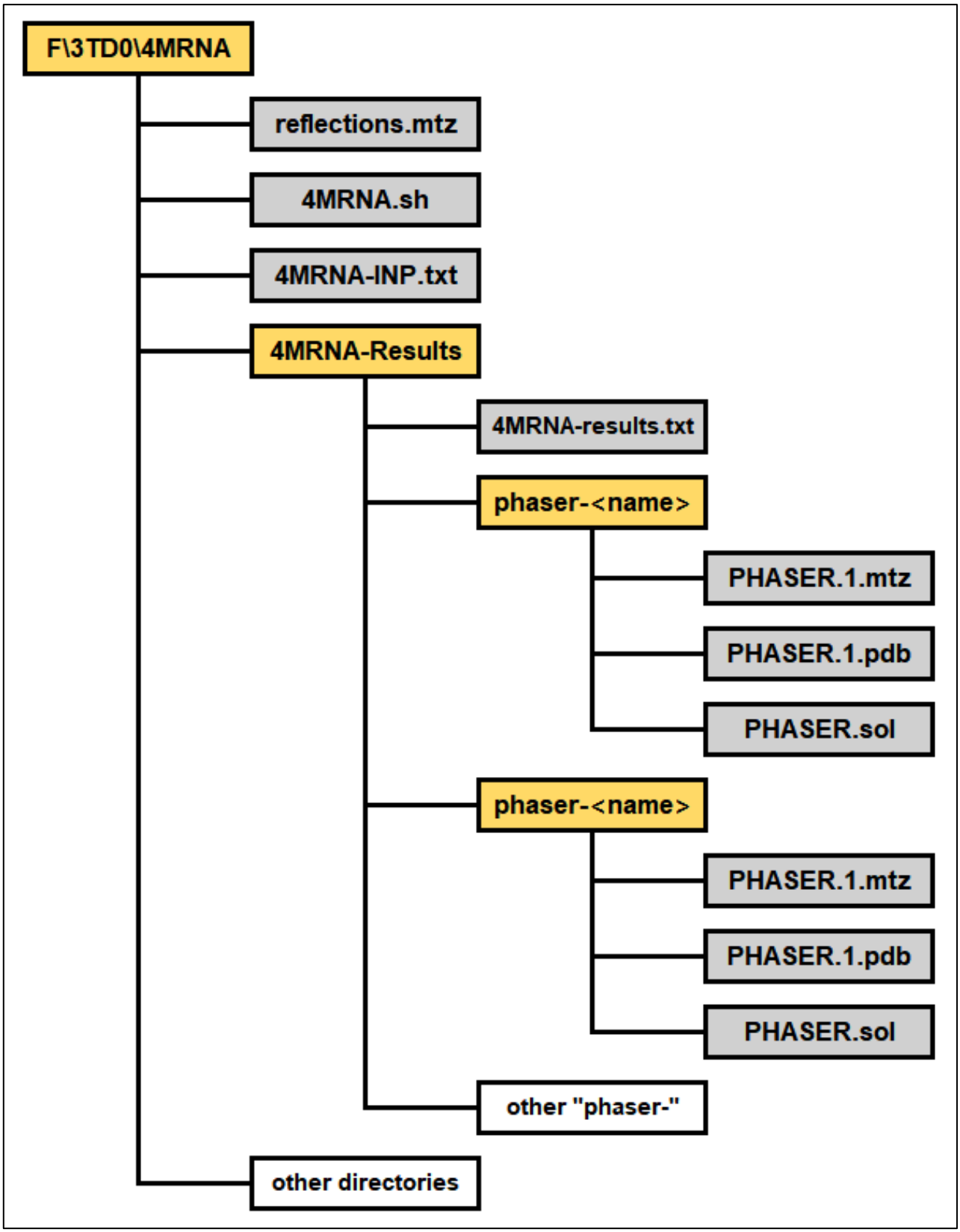
```
=== All finished ===
```



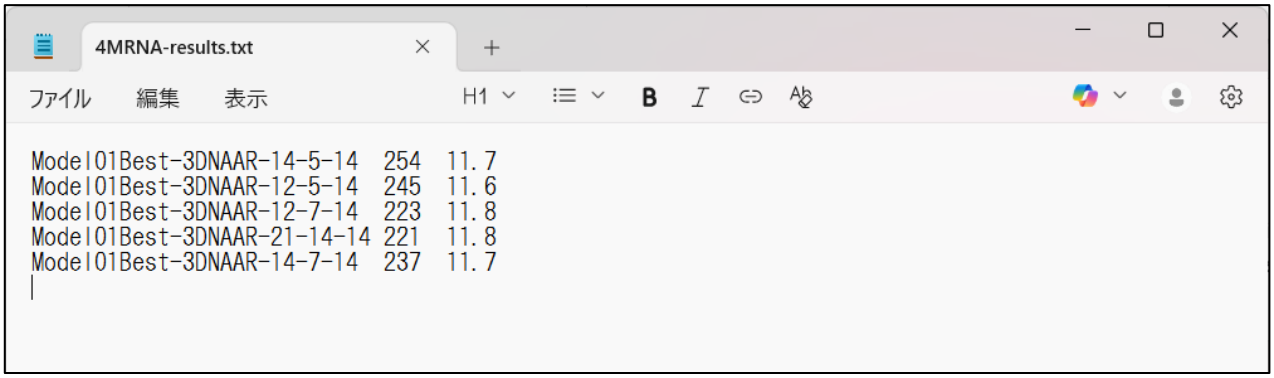
■ Inside the “4MRNA-Results” directory, there is a file named “**4MRNA-results.txt**” and up to seven subdirectories. Each subdirectory name begins with “**phaser-**” and contains the results of molecular replacement (MR).



- Inside each “phaser-” subdirectory, there is a structural file named “**PHASER.1.pdb**.” This file is a solution of molecular replacement, that is, a candidate for the correct structure.
- The following figure is an example of the final directory structure.



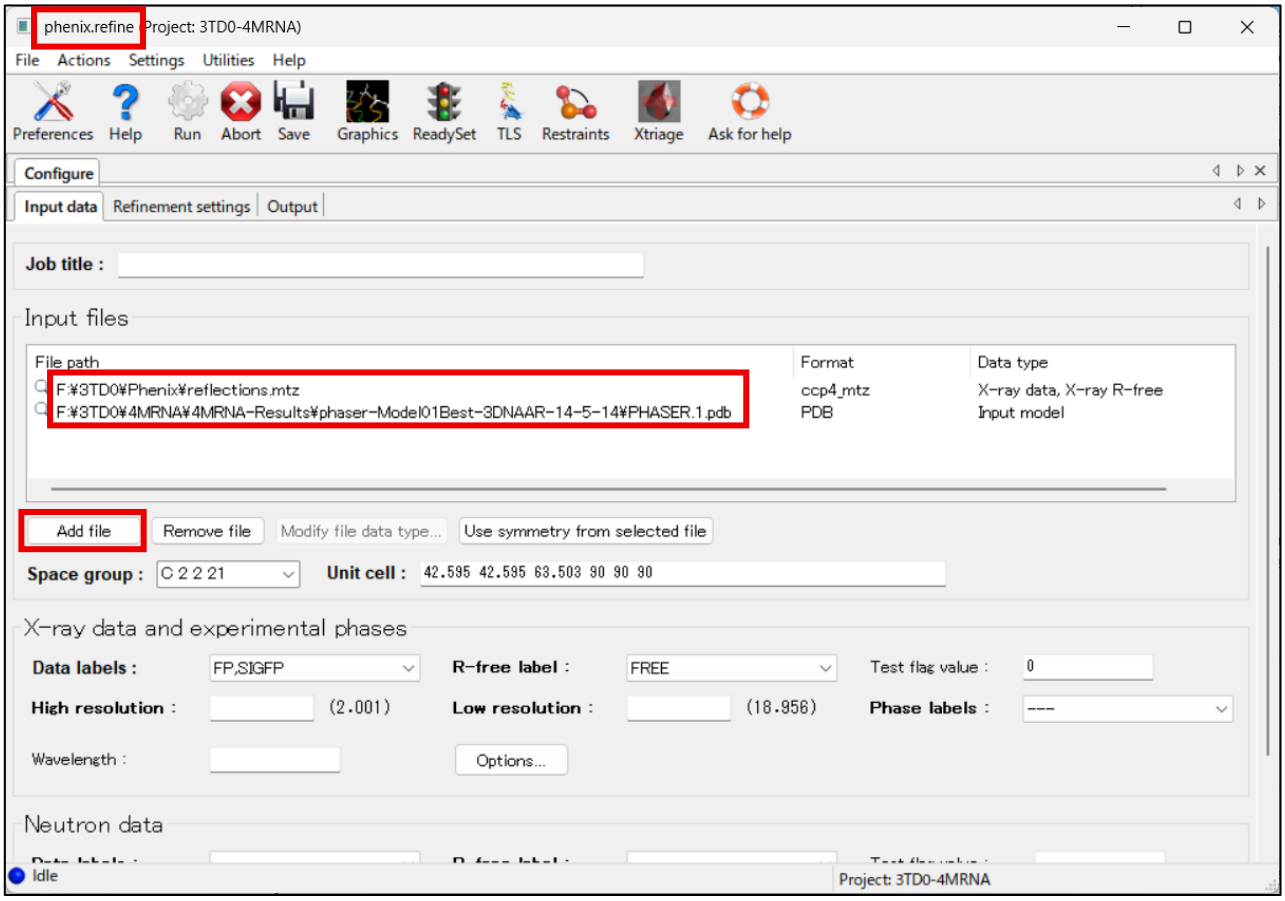
■ Each run of molecular replacement produces a solution, but it must be validated against the experimental data (electron density) to confirm its correctness. This validation is guided by two statistical values, **LLG** and **TFZ**. The program is designed to automatically select up to seven solutions with the most favorable statistics. These statistics are compiled in "**4MRNA-results.txt**." Each line in this file lists the ID of molecular replacement (i.e. the name of directory) , LLG, and TFZ in that order.



■ With reference to the statistics provided in "**4MRNA-results.txt**," open the structural file ("**PHASER.1.pdb**") together with the electron density map ("**PHASER.1.mtz**") in software like Coot. By overlaying them, decide which of the up to seven candidate solutions is the best.

■ After the optimal structure has been determined, it will be used as the initial structure for refinement.

[e.g.] For example, if the structure with the molecular replacement ID "Model01Best-3DNAAR-14-5-14" (i.e., "PHASER.1.pdb" within the directory "phaser-Model01Best-3DNAAR-14-5-14") is decided as the initial structure, you should add this file. In addition, add the reflection file ("reflections.mtz").



4-3. Supplementary notes

- Actually, for the `4MRNA` command to work, an input file called "**4MRNA-INP.txt**" is required.
- There are two ways to prepare the input file "4MRNA-INP.txt":
  1. Prepare the input file yourself before running the `4MRNA` command.
     ⇒ Please refer to the template below.
  2. **[Strongly recommended]** Do not prepare the input file in advance. Instead, answer the questions prompted by the `4MRNA` command, and the program will automatically create the input file.
     ⇒ Please refer to Section 4-2.
- The template of the input file "4MRNA-INP.txt" looks like the screenshot below. The following three pieces of information should be repeated for each model used in molecular replacement.

  -TYPE: This is the nucleic acid type. Choose from A-form DNA, B-form DNA, or A-form RNA.

  -SEQ: This is the sequence. Only the sequence of one strand of the duplex is required. The complementary strand is processed automatically by the program.

  -NUM: This means how many copies of the model to be searched in molecular replacement.

```
-TYPE1 A-DNA
-SEQ1 ACGTACGTACGT
-NUM1 1

-TYPE2 B-DNA
-SEQ2 ACGTACGT
-NUM2 1

-TYPE3 A-RNA
-SEQ3 ACGU
-NUM3 1
```

5. How to use the Customizable style

5-1. Very brief summary of the procedure

Access the website called “**Programming Code Generator for 4MRNA**” at the URL below. Usage instructions are provided on the website.
https://s-ando-biophysics.github.io/4MRNA/

5-2. Summary of the procedure

Step 1. Create a new directory on your computer for executing 4MRNA. After creating it, place your reflection file (.mtz) for molecular replacement into that directory.
Step 2. Access the website. https://s-ando-biophysics.github.io/4MRNA/
Step 3. Fill in the table on the website and click "**Generate codes (Customizable style)**" button.
Step 4. Click "**Download all codes**" button.
Step 5. Run all codes in the order indicated by the number at the beginning of each downloaded file.

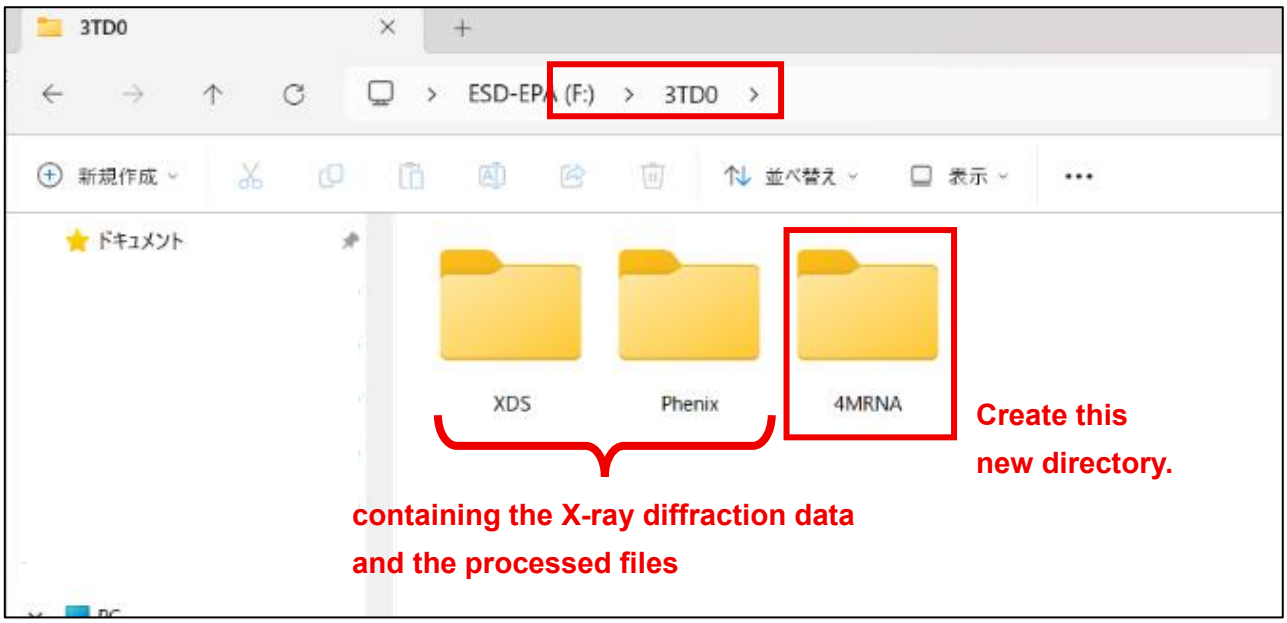
Type	How to run
Python (.py)	Open the code with a text editor such as Visual Studio Code and run them.
Shell (.sh)	Run the code by `bash <name>.sh` or `chmod +x <name>.sh && ./<name>.sh` in Ubuntu/Terminal.

Step 6. After all the codes have finished running, a directory named “**4MRNA-Results**” will be created, containing up to seven candidate solutions for molecular replacement.

5-3. Step-by-step instructions with screenshots

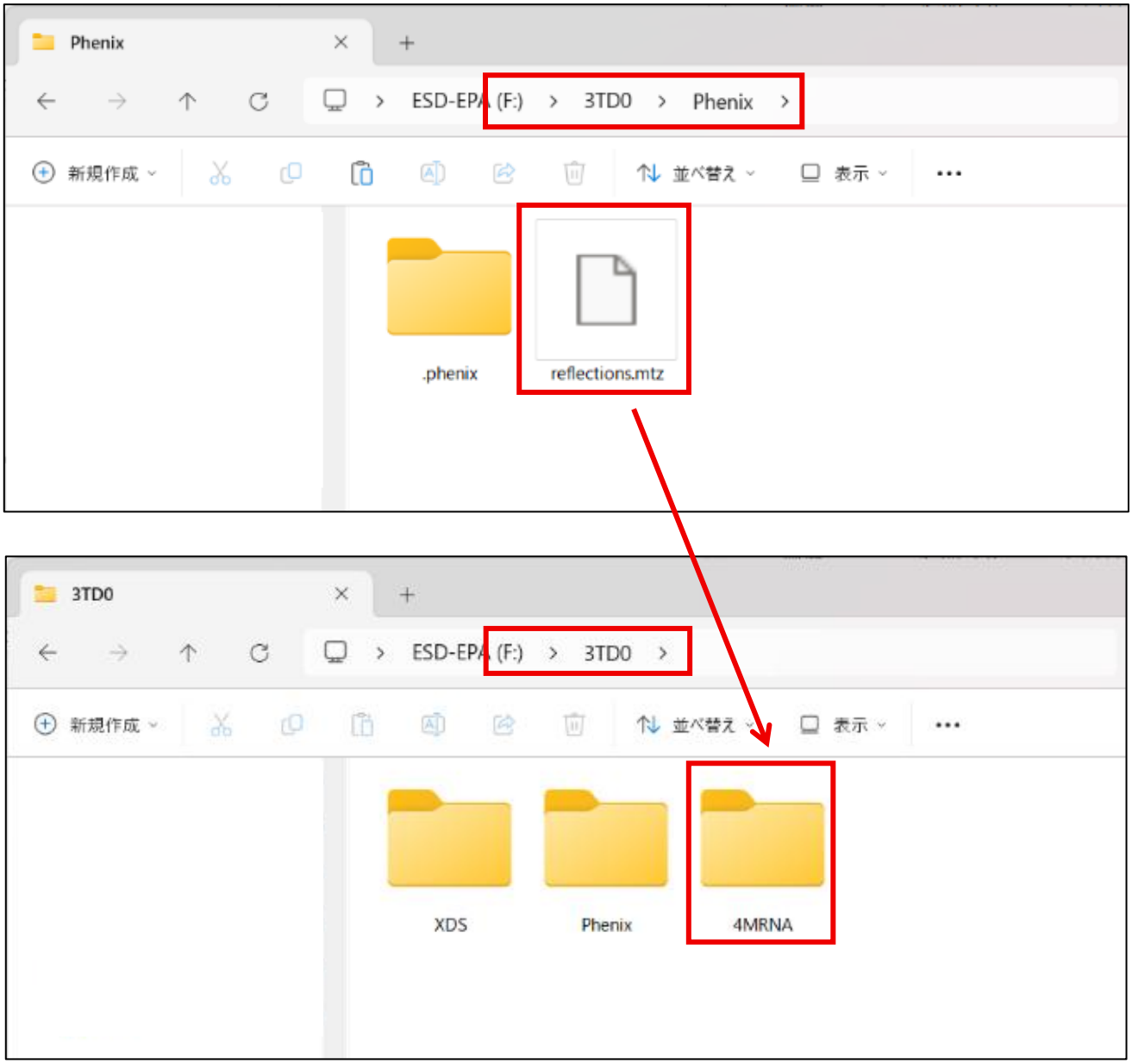
■ Create a new directory on your computer for executing 4MRNA.

[e.g.] Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a directory 3TD0 on a USB drive (F:). In that case, create a new directory named “4MRNA” inside “F:\3TD0”.

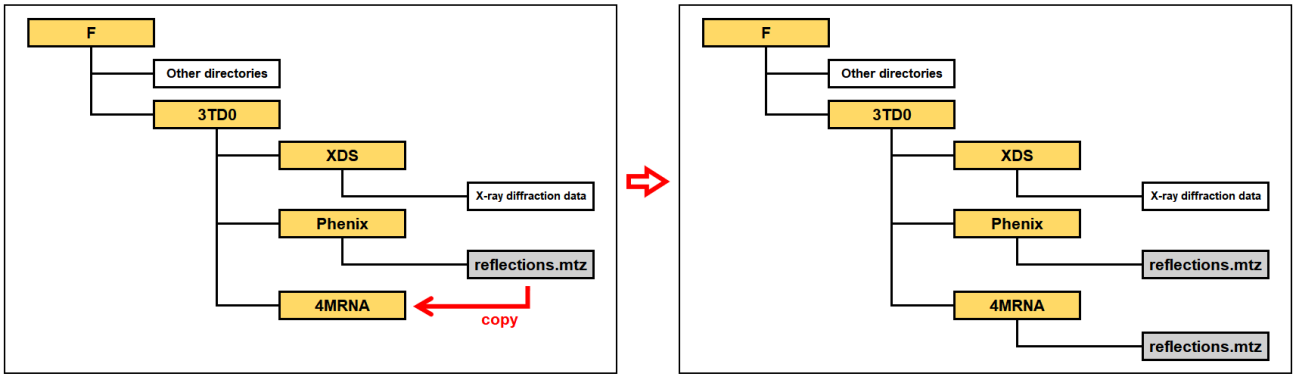


■ Place your reflection file (.mtz) for molecular replacement into the created directory.

[e.g.] When using the software Phenix, the reflection file is typically named “reflections.mtz”. If this file is in “F:\3TD0\Phenix”, copy it, and paste to “F:\3TD0\4MRNA”.



■ The directory structure after these operations is as illustrated in the example below.

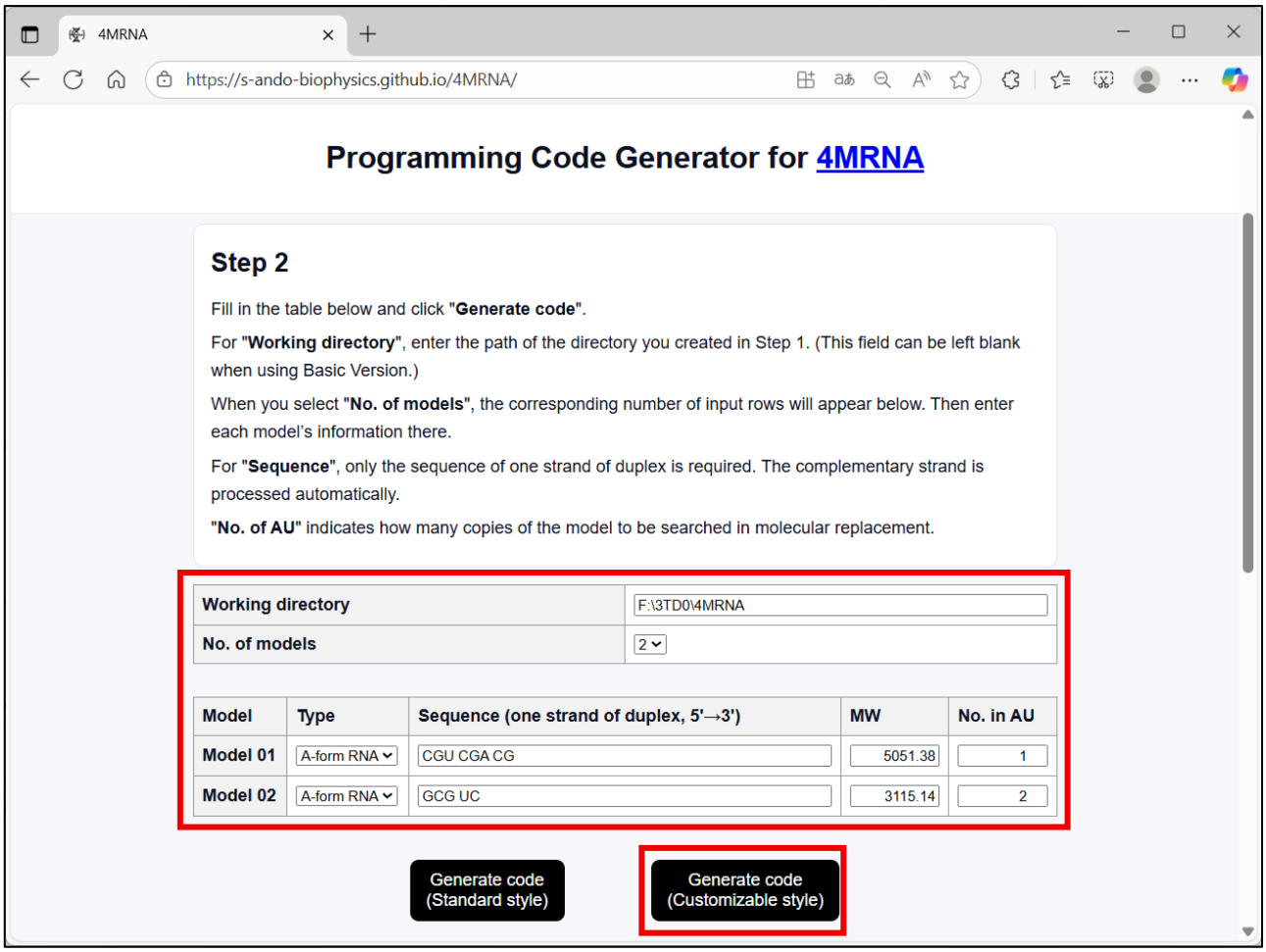


- Access the website called “**Programming Code Generator for 4MRNA**” at the following URL.
  https://s-ando-biophysics.github.io/4MRNA/
- Fill in the table on the website. The contents of the table are as follows.

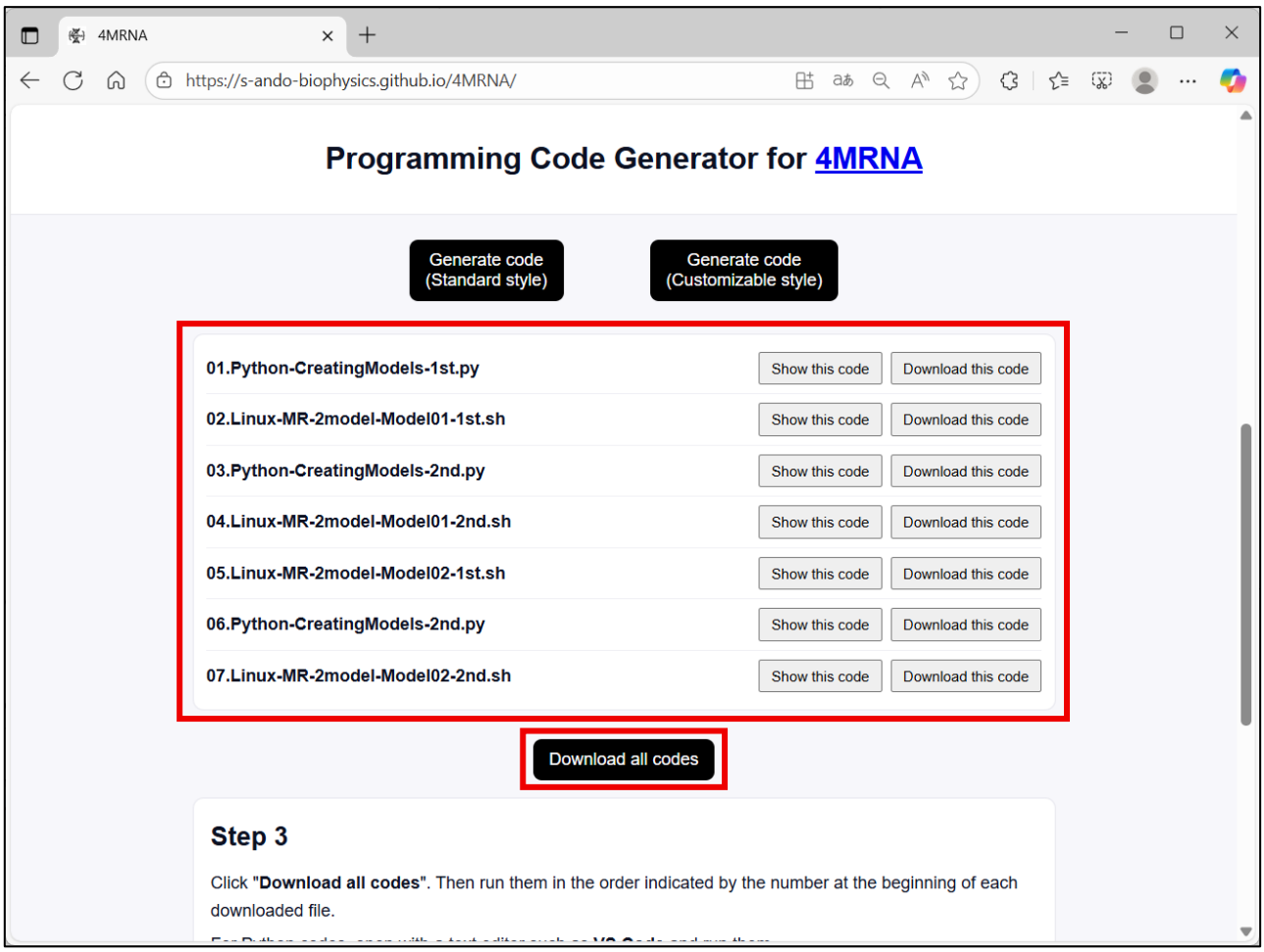
Working directory		Enter the path of the directory you created at the beginning.
No. of models		How many different duplexes do you want to use as search models for molecular replacement?
Each model	**Type**	Enter the nucleic acid type. Choose from “A-form DNA”, “B-form DNA”, or “A-form RNA”.
	Sequence	Enter the sequence of each model. Only the sequence of one strand of the duplex is required. The complementary strand is processed automatically by the program.
	MW	This value is automatically calculated from the sequence (including the complementary strand), so no input is required.
	No. in AU	How many copies of the model do you want to search for in molecular replacement?

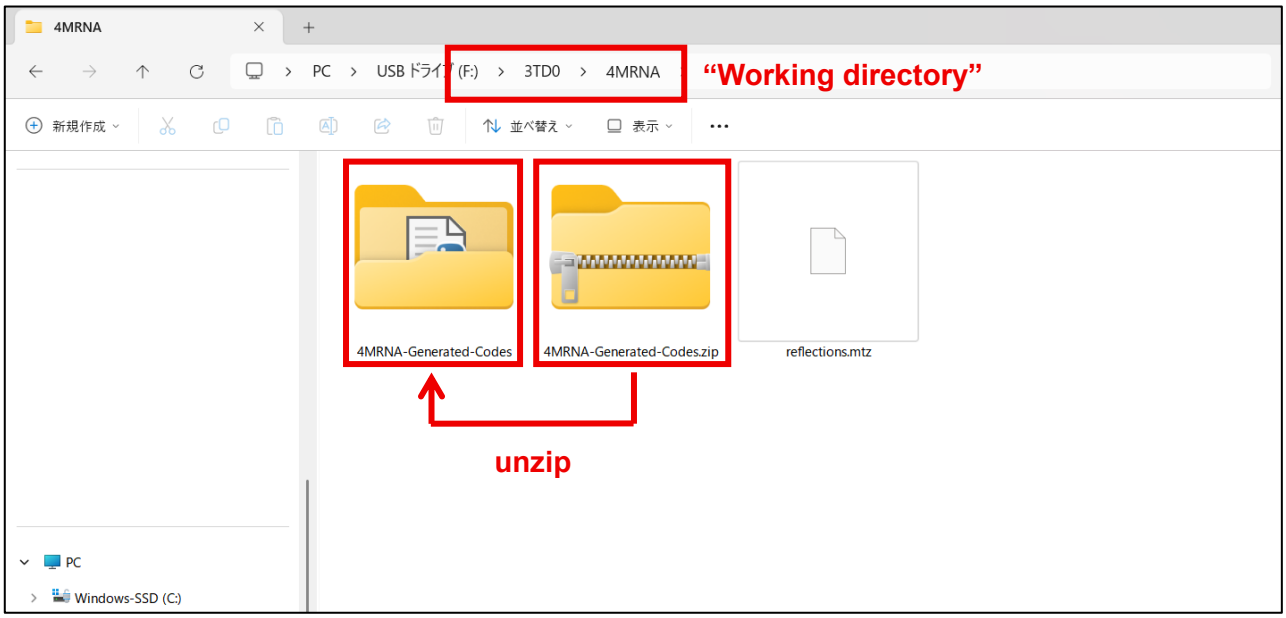
- After entering all the items, and then click "**Generate codes (Customizable style)**" button.

[e.g.] The figure below shows an example in which the directory created at the beginning has the path “F:\3TD0\4MRNA,” and molecular replacement is performed using two different RNAs as models. In this example, during molecular replacement, one copy of the first model and two copies of the second model are searched for in the asymmetric unit of the crystal.

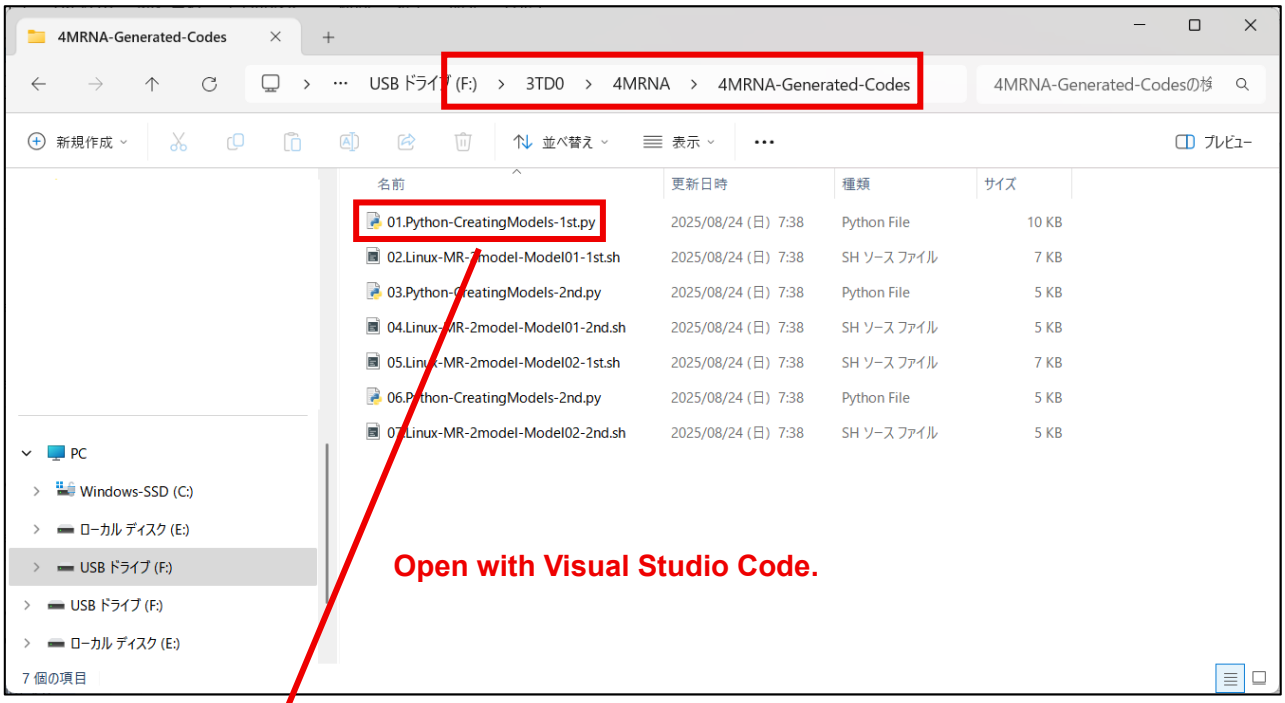


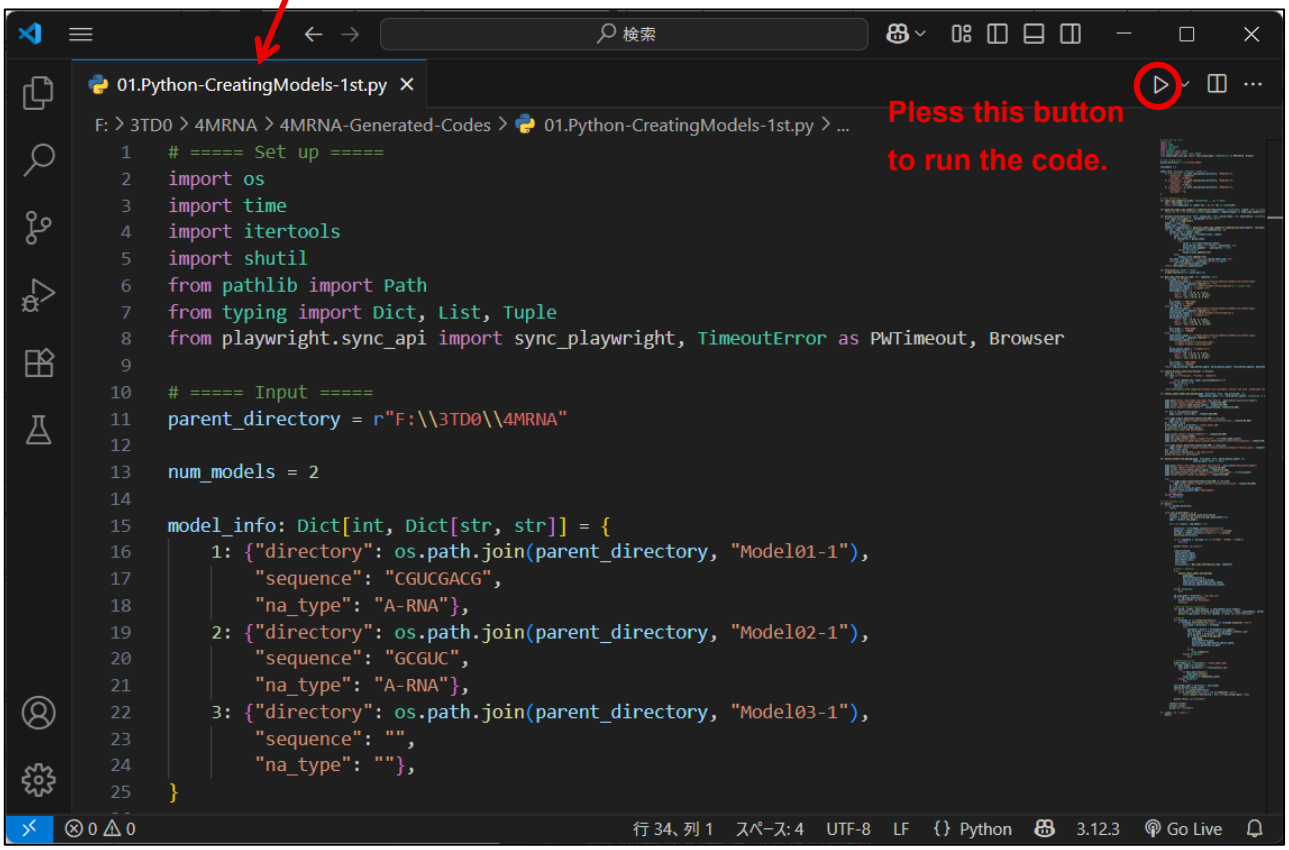
- When you click the “Generate codes (Customizable style)” button, codes tailored to your sample information will appear on the screen. Download them by clicking “**Download all codes**” button. It is recommended that the downloaded files be saved in the directory created at the beginning (i.e. “Working directory”).
- The files are downloaded as a zip file named “**4MRNA-Generated-Codes.zip**”, so please unzip it.





- After unzipping, the directory “**4MRNA-Generated-Codes**” contains Python codes and Shell scripts.
- These codes include the information you entered in the table on the website.
- Basically, you only need to run the codes in the order indicated by the numbers at the beginning of each file, and the entire 4MRNA workflow will be completed.
- However, it is also possible to review and modify the contents of each file if necessary.
- For the **Python** codes (.py), it is recommended to open with a text editor such as Visual Studio Code and run it.

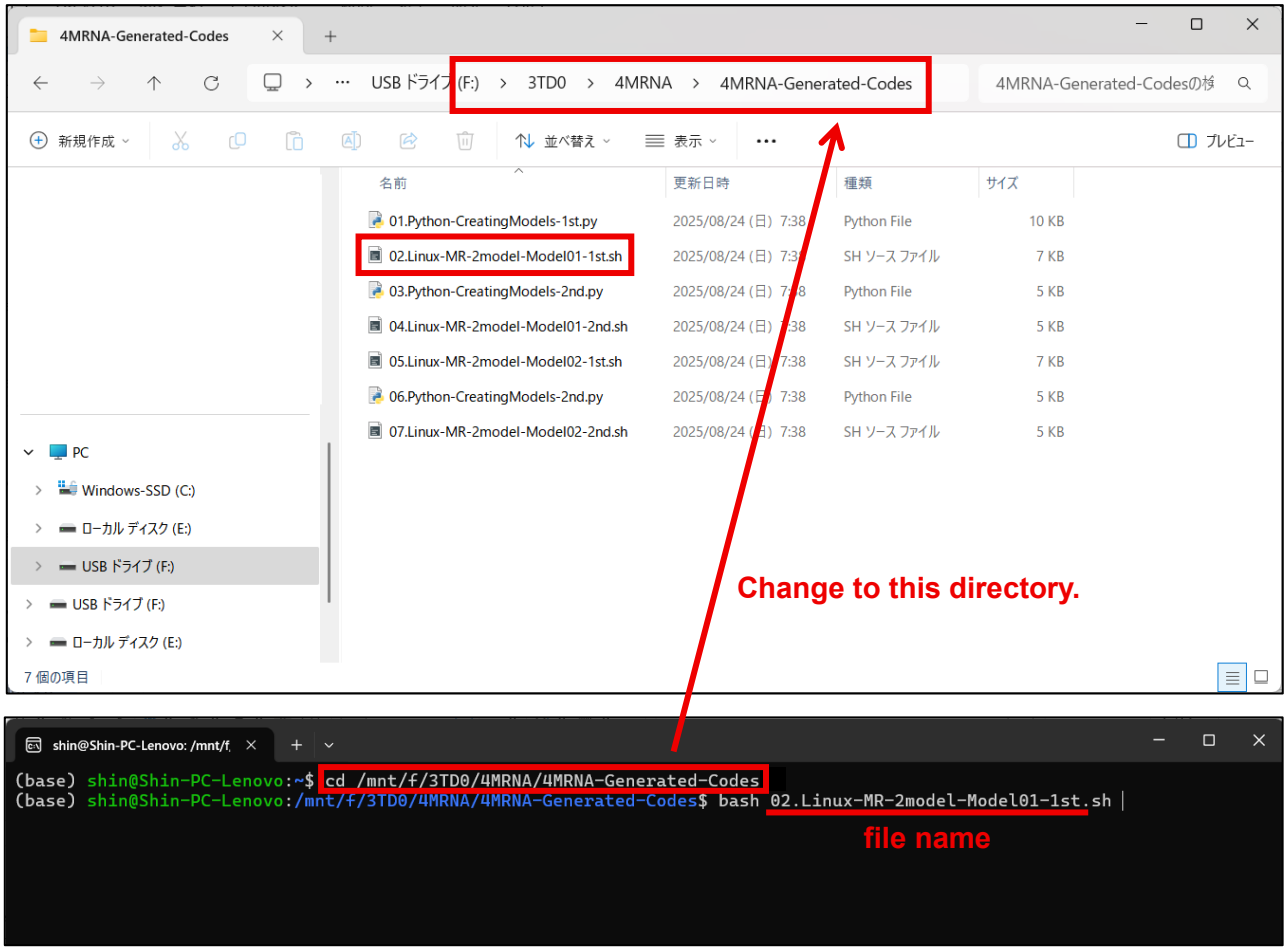




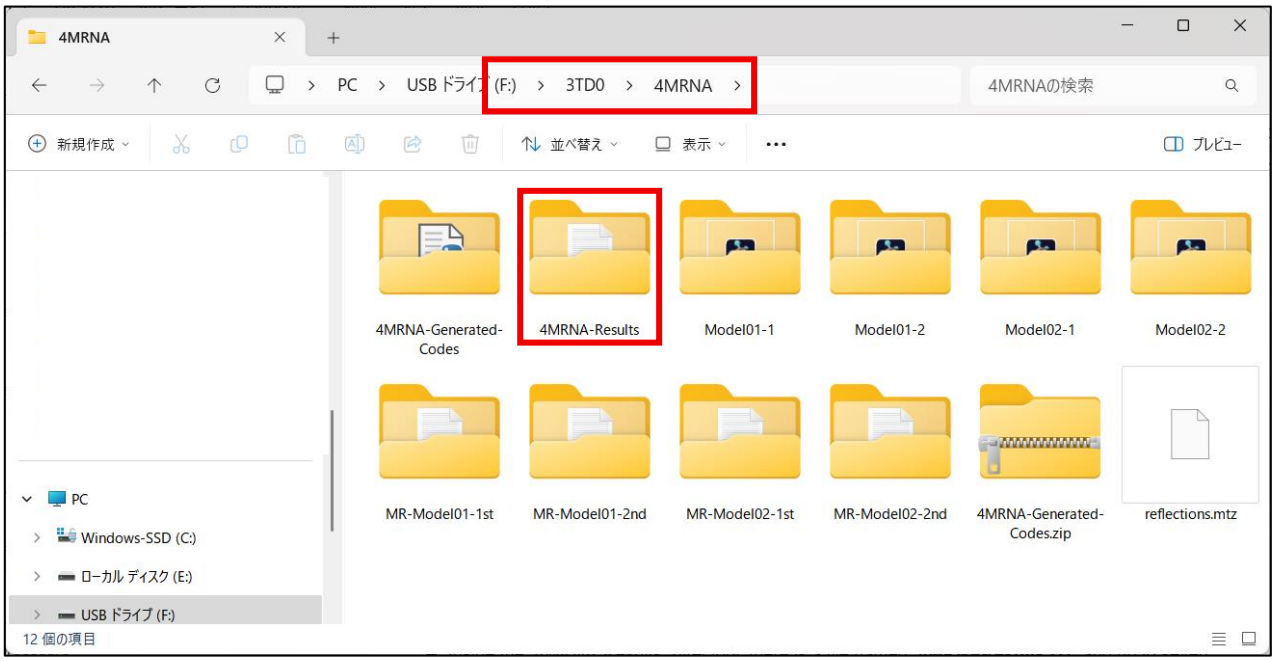
■ For the **Shell** scripts (.sh), open Ubuntu or Terminal, change to the directory "4MRNA-Generated-Codes", and run the code as follows.

`bash <file name>.sh` or `chmod +x <file name>.sh && ./<file name>.sh`

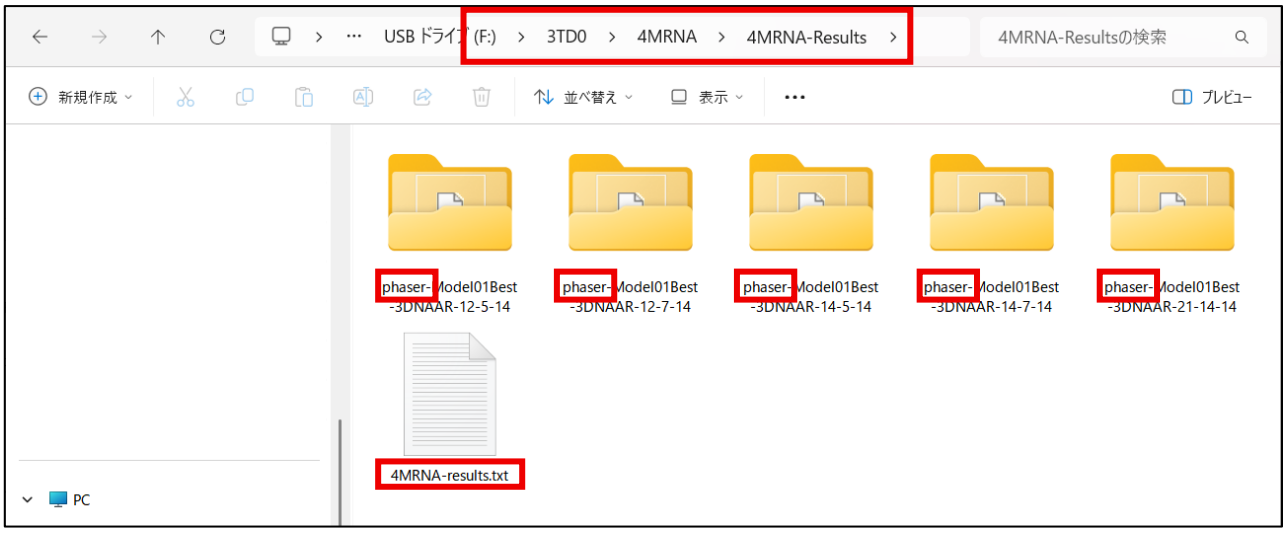
■ Of course, you can also run it by copying the entire contents and pasting them into Ubuntu/Terminal.



■ After all the codes have finished running, a directory named "**4MRNA-Results**" will be created

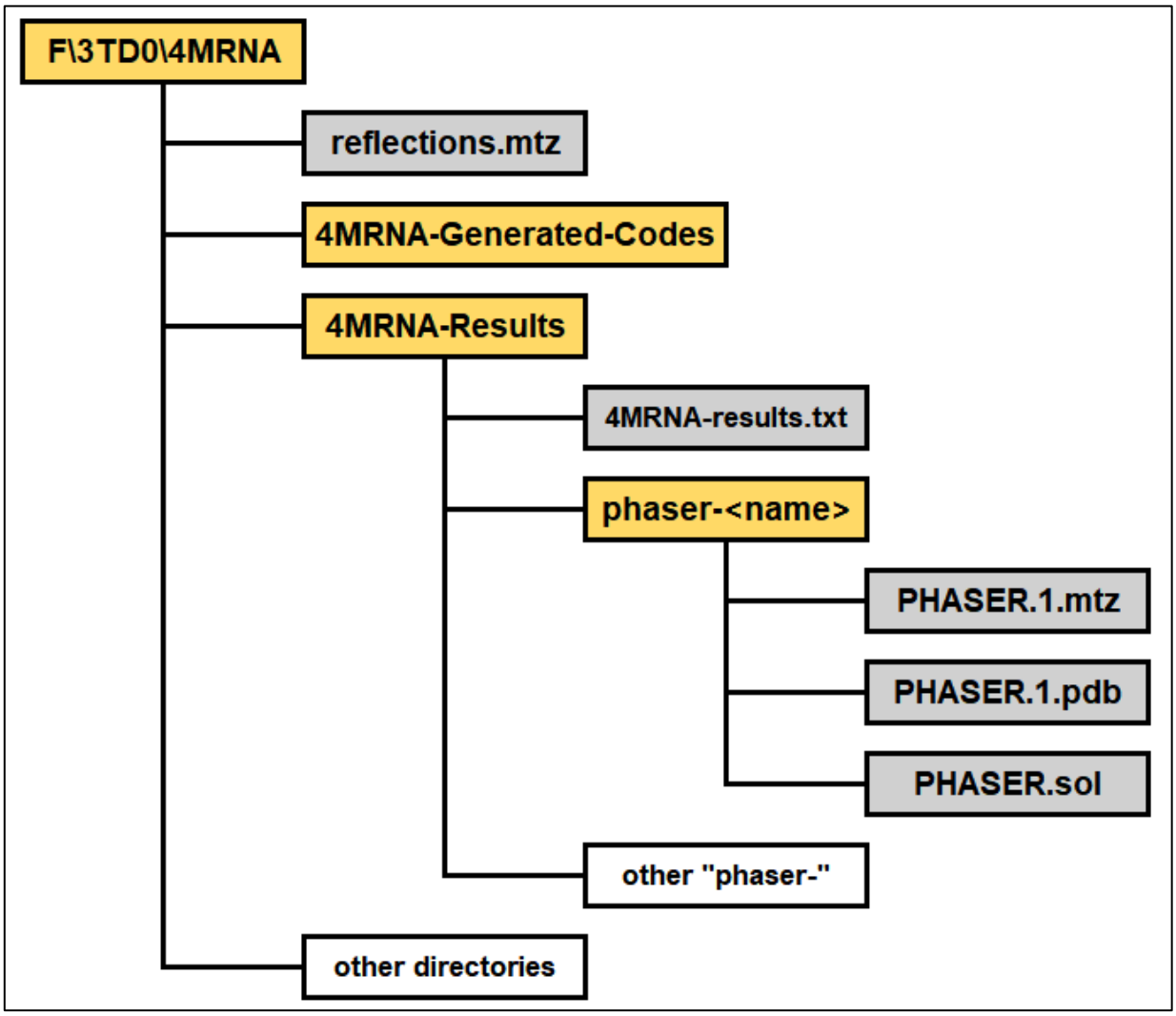


■ Inside the “4MRNA-Results” directory, there is a file named “**4MRNA-results.txt**” and up to seven subdirectories. Each subdirectory name begins with “**phaser-**” and contains the results of molecular replacement (MR).

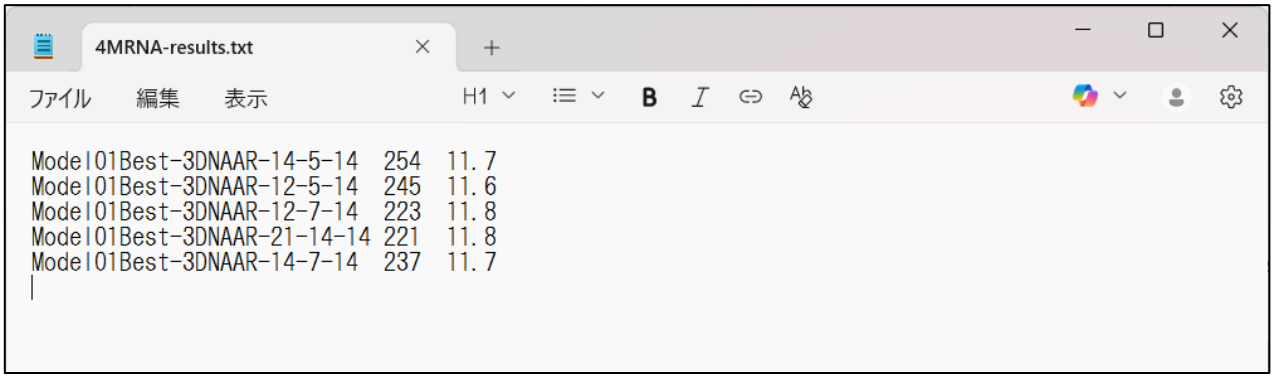


■ Inside each “phaser-” subdirectory, there is a structural file named “**PHASER.1.pdb**.” This file is a solution of molecular replacement, that is, a candidate for the correct structure.

■ The following figure is an example of the final directory structure.



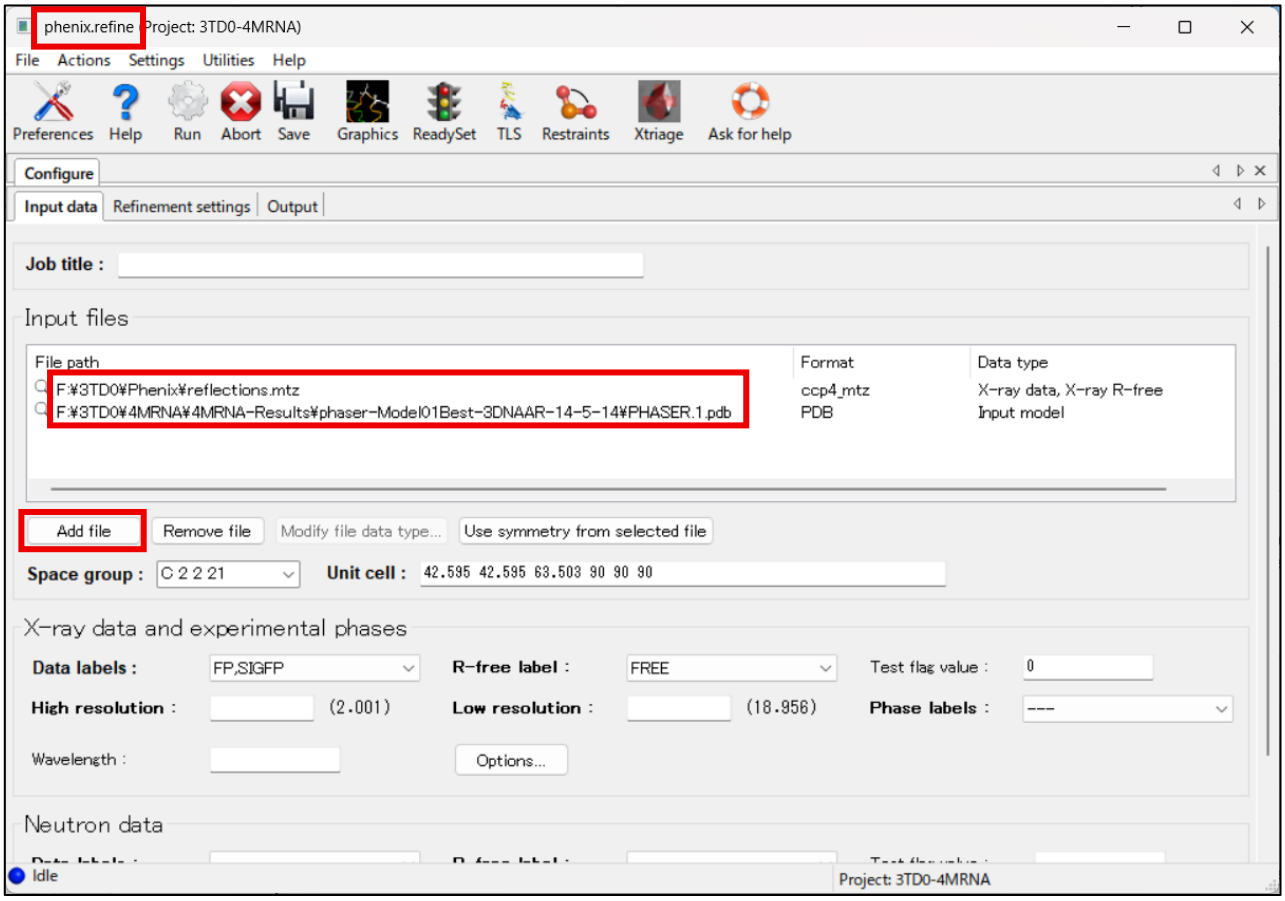
■ Each run of molecular replacement produces a solution, but it must be validated against the experimental data (electron density) to confirm its correctness. This validation is guided by two statistical values, **LLG** and **TFZ**. The program is designed to automatically select up to seven solutions with the most favorable statistics. These statistics are compiled in "**4MRNA-results.txt**." Each line in this file lists the ID of molecular replacement (i.e. the name of directory) , LLG, and TFZ in that order.



■ With reference to the statistics provided in "**4MRNA-results.txt**," open the structural file ("**PHASER.1.pdb**") together with the electron density map ("**PHASER.1.mtz**") in software like Coot. By overlaying them, decide which of the up to seven candidate solutions is the best.

■ After the optimal structure has been determined, it will be used as the initial structure for refinement.

[e.g.] For example, if the structure with the molecular replacement ID "Model01Best-3DNAAR-14-5-14" (i.e., "PHASER.1.pdb" within the directory "phaser-Model01Best-3DNAAR-14-5-14") is decided as the initial structure, you should add this file. In addition, add the reflection file ("reflections.mtz").



6. Notes

6-1. Supported environment

- **Operating system (OS)**: Windows [*], macOS, Linux (Rocky Linux)
- **Browser**: Google Chrome, Microsoft Edge, Firefox, Safari

[*] Shell scripts were executed in Windows Subsystem for Linux (WSL) with Ubuntu.

6-2. License information

- **4MRNA itself is distributed under the MIT License.** This means that it can be used completely freely, provided that the copyright notice is included.
- However, please pay attention to the licenses of the dependencies. 4MRNA relies on several external resources and libraries, including Web 3DNA, CCP4/Phaser, Pandas (Python library), and Playwright (Python library), but these components are not included in the distribution of 4MRNA. Users are responsible for installing them separately and complying with their respective licenses.
- A summary of these dependencies is provided in "DEPENDENCIES.txt" on GitHub.
- In practical terms, the summary is as follows. All software is available free of charge for academic and non-profit purposes under the appropriate agreements, whereas commercial use requires careful attention to the licensing terms of Phaser (CCP4).

6-3. GitHub links

- **4MRNA (repository)**: https://github.com/S-Ando-Biophysics/4MRNA
- Standard style of 4MRNA (release): https://github.com/S-Ando-Biophysics/4MRNA/releases/latest
- Install style of 4MRNA (repository): https://github.com/S-Ando-Biophysics/4MRNA-Install

6-4. Citation

The paper is in preparation. Once it is published, please cite that paper when using 4MRNA.

7. References

[1] Bernard, C., Postic, G., Ghannay, S., & Tahi, F. (2025). Has AlphaFold3 achieved success for RNA?. *Acta crystallographica. Section D, Structural biology*, *81*(Pt 2), 49–62.

[2] Kwon D. (2025). RNA function follows form - why is it so hard to predict?. *Nature*, *639*(8056), 1106–1108.

[3] Kondo, J., Urzhumtseva, L., & Urzhumtsev, A. (2008). Patterson-guided ab initio analysis of structures with helical symmetry. *Acta crystallographica. Section D, Biological crystallography*, *64*(Pt 10), 1078–1091.

[4] Li, S., Olson, W. K., & Lu, X. J. (2019). Web 3DNA 2.0 for the analysis, visualization, and modeling of 3D nucleic acid structures. *Nucleic acids research*, 47(W1), W26–W34.